

Relative Valuation Using Market Comparables

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Relative Valuation Using Market Comparables

Chapter Overview

In this chapter, we discuss *relative valuation*, or valuation using *market comparables*—a technique that is often used to value businesses, business units, and other major investments. We introduce the use of relative valuation in the context of a valuation problem faced by virtually everyone—the valuation of residential real estate. Although this illustration is quite simple, it illustrates the basic tenets of the use of relative valuation. Relative valuation criteria are often viewed as substitutes for the discounted cash flow (DCF) analysis we described in earlier chapters. As the discussion in this chapter illustrates, the DCF approach can be viewed as the conceptual basis for most relative valuation criteria, and the two approaches should be seen as complements.

1 INTRODUCTION

Relative valuation uses market prices from observed transactions to impute the value of a firm or investment opportunity. If you are considering the sale of your home, you might estimate the appropriate asking price by looking at **market comparables**. For example, you might calculate the ratio of prices from recent sales in the neighborhood to the square footage of the homes to calculate a price per square foot and then multiply this number by the number of square feet in your house to get an estimate of what the home should sell for. For income-producing investments, analysts consider additional ratios, which include market values relative to earnings, cash flow, sales, or the book value of recent transactions.

Up to this point, we have estimated the value of an investment by discounting cash flows, which requires that the analyst develop a forecast of expected future cash flows as well as an appropriate discount rate. As we will illustrate in this chapter, valuing investments using *market comparables* (or *comps*) essentially solves the same problem using different information. Specifically, instead of trying to estimate future cash flows, this approach uses information from the market prices of comparable transactions. The critical assumption that underlies this approach to valuation is that the “comparable” transactions are truly comparable to the investment being evaluated.

Even though the science behind relative valuation is relatively easy to grasp, the successful application of this valuation methodology requires considerable skill. To help

you develop that skill, we have organized the chapter as follows: In Section 2, we introduce relative valuation using examples from both residential and commercial real estate. Here we present the essential elements of the comparables valuation process, and we demonstrate the link between relative valuation multiples and DCF valuation. In Section 3, we discuss the use of comparables to value an entire business or enterprise. We learn that one of the most popular valuation ratios is the ratio of enterprise value to EBITDA (earnings before interest, taxes, depreciation, and amortization). In Section 4, we illustrate the use of the price–earnings multiple to value a firm’s common equity, which is arguably the most common use of relative valuation. Section 5 discusses the valuation of a firm’s equity when company shares are sold to the public for the first time (i.e., an initial public offering). In Section 6, we discuss a number of critical issues that arise in the use of relative valuation using market-based comparables. Finally, Section 7 contains summary comments.

2 VALUATION USING COMPARABLES

Relative valuation using market comparables involves a four-step process. To illustrate the steps, consider the valuation of a service station in Tulsa, Oklahoma.

Step 1: Identify similar or comparable investments and recent market prices for each.

This is the most critical step in the process; the quality of the resulting estimate of value depends on how carefully matched the comparables are to the investment being valued.

Example: Within the last six months, two other service stations in Tulsa have sold. Sales prices and other pertinent information for the sales are available for analysis.

Step 2: Calculate a “valuation metric” for use in valuing the asset. This involves identifying a key attribute of the investment being valued and dividing it into the market prices of the comparables to compute a valuation ratio. Examples of attributes that are commonly used include earnings per share (when valuing common stock), the number of square feet (when valuing a home or building), and earnings before interest, taxes, depreciation, and amortization (when valuing a business enterprise). Once we have calculated the valuation ratios for the market comps, we typically average them to calculate the *valuation ratio* used in valuing the asset.

Example: The valuation of businesses like the Tulsa service station typically uses a valuation ratio based on some definition of earnings. For illustration purposes, we will use the operating earnings and define the valuation ratio as the sale or transaction price divided by the station’s net operating earnings. For the two service station sales we have identified as market comps, we have the following information:

Station	Sale Price	Net Operating Earnings	Valuation Ratio
A	\$ 900,000	\$450,000	2
B	\$1,200,000	\$100,000	12
Average valuation ratio			7

Step 3: Calculate an initial estimate of value. Multiply the valuation ratio (typically the average of a set of market comps) by the key “attribute” of the investment whose value is being estimated.

Example: The operating earnings of the Tulsa service station were \$150,000 for the most recent year. Applying the average valuation ratio of 7 to these earnings produces an initial value estimate of \$1,050,000.

Step 4: Refine or tailor your initial valuation estimate to the specific characteristics of the investment. In “refining the estimate,” we analyze and, if need be, adjust the valuation ratio to reflect the peculiarities of the investment being valued.

Example: The immediate concern we have about the initial estimate for the Tulsa service station relates to the dramatic disparity in valuation ratios for the two comparable sales. Why did Station B sell for a multiple of 12 times operating earnings, while Station A commanded a multiple of only 2? At this point, we will want to make sure that the numbers used in arriving at the comparable valuation ratios are reasonable. For example, it may be that Station B actually generated operating earnings of only \$100,000 in the year of the analysis; however, in the year of the sale, the station’s earnings were depressed by the partial closure of its main entrance for six months due to road improvements. If we normalized the firm’s operating earnings to reflect our estimate of what operating earnings would have been without the construction, Station B would have had operating earnings of \$200,000. With the normalized earnings figure, the valuation ratio for Station B would have been only 6 (not 12). Using the revised valuation ratio for Station B and assuming that Station A was not similarly affected, we get an average valuation ratio of 4.0 and a value estimate for the station being analyzed of only \$600,000 ($4.0 \times \$150,000$), which is dramatically lower than the initial estimate.

To gain additional perspective on the issues raised in the service station valuation example, we include two real estate valuation examples. The first entails an analysis of a residence, and the second is a more complicated commercial real estate example.

Valuing Residential Real Estate Using Comparables

Anyone who has ever sold or bought a house through a real estate agent has experienced the use of relative valuation based on the method of comparables. The standard valuation approach for residential housing is to look for the price per square foot at which similar properties have recently sold. Just like our service station example, the valuation of a house begins by looking for comparable sale transactions. Perhaps the best way to gain an appreciation for the nuances of this methodology is by the use of a simple example.

Example: Consider the valuation of a home with 3,581 square feet of floor space, located in McGregor, Texas (five miles east of Former President George W. Bush’s ranch in Crawford, Texas). The house is only one year old, is situated on an oversized lot, and has a swimming pool. The owner has set an asking price of \$385,000 but has indicated a willingness to be flexible. Our problem is to estimate what the house should sell for, given the information on two recent market transactions in the same neighborhood.

Relative Valuation Using Market Comparables

The home is located in a popular area where recent sales have been recorded. The subdivision in which the home sits is less than five years old, so all the homes are of approximately the same age and condition. Two sales have occurred in the immediate neighborhood over the previous six months. Details on these sales are found below:

	Comp #1	Comp #2
Sale price	\$330,000	\$323,000
Square footage	3,556	4,143
Selling price/sq. ft.	\$ 92.80	\$ 77.96
Time on the market	97 days	32 days

We can use the selling price per square foot for the two comps to estimate the value of the home, by multiplying the home's square footage of 3,581 by the corresponding selling prices per square foot for the comps.

Valuation Using Comps	Comp #1	Comp #2	Average
Comparable sq. ft. selling price	\$ 92.80	\$ 77.96	\$ 85.38
Square footage of house being valued	3,581	3,581	3,581
Estimated value	\$332,317	\$279,175	\$305,746

The estimates of the value of the home range from \$279,175, based on the sale price of Comp #2, to \$332,317, based on the sale price of Comp #1. The average of the two price estimates is \$305,746.

Up to now, we have not taken into account two distinctive features of the house we are valuing. Specifically, it is located on a lot that is somewhat larger than the average lot in the subdivision, which may be worth \$15,000–\$20,000 more than the average lot. In addition, the home has a new swimming pool, which cost about \$30,000 to build.

Valuation Using Comps	Comp #1	Comp #2	Average
Estimated value	\$332,317	\$279,175	\$305,746
Adjustments			
Plus lot premium	20,000	20,000	20,000
Plus pool	15,000	15,000	15,000
Estimated market value	\$367,317	\$314,175	\$340,746

To adjust for these two differences, we add a lot-size premium of \$20,000 plus half the cost of installing the pool, or \$15,000. (Not everyone wants a swimming pool, and typically pools increase the price of the home by about half their cost when the home is sold.) This raises our average estimated value of the house to \$340,746.

Is the house worth the estimated \$340,746? The answer is an emphatic *maybe*. The reason for hesitating is that there are always intangibles that influence the value of a house (or a company, for that matter) that are not captured in the pricing of similar

properties (comps). For example, it is not uncommon for houses that are built on premium lots to have special amenities or premium features. Although we do not know whether this is the case here, it is important that we consider the unique attributes of the house before arriving at our final estimate of its value.

This simple housing valuation example illustrates three key points concerning the use of the method of comparables.

- Point #1: *Identification of appropriate comps is paramount.* The valuation estimate is only as good as the set of comparables selected to determine the valuation multiple. Consequently, a real effort should be made to identify appropriate comparables.
- Point #2: *The initial estimate must be tailored to the investment's specific attributes.* The initial value estimate is only the beginning of the analysis. Even carefully selected comparables are almost always imperfect replicas of the investment being evaluated. As a result, the method of comparables requires that the analyst make a number of adjustments. In this case, we were able to make adjustments for known differences between the house we are valuing and the comparison homes. Even after this process is complete, however, there are often other intangible differences that can lead to differences of opinion as to the actual value of the house. When a business or a large investment project is valued, these types of adjustments can be extremely important.
- Point #3: *The specific metric used as the basis for the valuation can vary from one application to another.* In our residential real estate valuation example, we used square footage as the comparable “feature,” yet we used operating earnings in the service station example. In general, analysts use multiples of cash flows or some measure of earnings when reliable measures are available, but rely on less direct measures of an investment’s earnings potential in other cases. In the case of a residential home, the owner receives “housing services,” which can be viewed as his or her earnings from the property. When we use square footage (as opposed to cash flows) as the basis for comparing houses, we are assuming that the flow of housing services is directly related to the size of the house. Similarly, when valuing cable companies, analysts often use the number of cable subscribers as the key valuation metric. The important thing is that we select a valuation metric that is closely related to the investment’s ability to generate cash flows or other benefits.

Valuing Commercial Real Estate

The residential real estate example involving the house in McGregor, Texas, should provide some feel for how to use the market comparables approach. Let’s now consider commercial real estate. Although commercial real estate is often valued with the same multiples or comparables as residential real estate, analysts generally value commercial real estate by evaluating cash flow ratios as well as prices per square foot. As we show below, the cash flow valuation multiples that are used in practice have a direct relationship to the DCF approach that we discussed in the earlier chapters.

Valuation Ratios (Multiples) and DCF Valuation

To illustrate the connection between DCF valuation and relative valuation, let’s consider the problem of valuing a perpetual cash flow stream of \$100 per year, where the

rate used to discount the cash flows is 20%. The present value of this cash flow stream is calculated as follows:¹

$$\text{Value} = \frac{\text{Cash Flows}}{\text{Discount Rate } (k)} \quad (1)$$

Substituting into Equation 1, we estimate a value of \$500, i.e.,

$$\text{Value} = \frac{\$100}{.20} = \$500$$

With a slight rearrangement of terms, we can express the DCF valuation as the product of the annual cash flow of \$100 and a multiple, i.e.,

$$\text{Value} = \$100 \times \left(\frac{1}{.20} \right) = \$100 \times 5 = \$500$$

In this form, we recognize that the multiplier used to value the stream of \$100 cash flows is $(1/.20) = 5$. It is customary when valuing commercial real estate to define what is known as a capitalization or “cap” rate. The **capitalization rate** is the reciprocal of the multiple used to value the property. In this instance, the cap rate is $1/5 = .20$, which in this example (a level perpetuity cash flow stream) is simply the discount rate. However, the cap rate is not always the discount rate; indeed, it is somewhat less than the discount rate when cash flows are expected to grow, and it exceeds the discount rate when cash flows are expected to shrink or decline over time.

To illustrate how the cap rate can differ from the discount rate, let’s consider the case where the \$100 cash flow grows at a rate of 5% per year forever and the discount rate is 20%. The DCF valuation equation (Equation 2) used in this situation is known as the **Gordon growth model**, i.e.,²

¹ The present value of a level perpetuity with periodic cash flow X and discount rate k can be shown to be equal to X/k . This result is well known in finance and constitutes the sum of the following geometric progression:

$$\sum_{t=1}^{\infty} \frac{X}{(1+k)^t} = X \sum_{t=1}^{\infty} \frac{1}{(1+k)^t} = X \left(\frac{1}{k} \right)$$

This analytical expression has a very intuitive basis. Think of the following situation: You want to generate a cash flow of X each year forever. How much would you have to put in a bank if the interest rate paid is k ? The answer is X/k , because the interest you earn each year is $k(X/k)$ —exactly X . Thus, investing an amount X/k is equivalent to generating a cash flow of X each year forever.

² The Gordon growth model is simply the sum of the following geometric progression:

$$\sum_{t=1}^{\infty} \frac{X_0 (1+g)^t}{(1+k)^t} = X_0 \sum_{t=1}^{\infty} \frac{(1+g)^t}{(1+k)^t},$$

which can be reduced to

$$X_0 \left(\frac{1+g}{k-g} \right) \text{ or } X_1 \left(\frac{1}{k-g} \right)$$

where $k > g$ and X_1 is equal to $X_0(1+g)$.

$$\text{Value} = \frac{\text{Cash Flow}_0 [1 + \text{Growth Rate } (g)]}{\text{Discount Rate } (k) - \text{Growth Rate } (g)} \quad (2)$$

Substituting into Equation 2 produces an estimate of value equal to \$700, i.e.,

$$\text{Value} = \frac{\$100 (1 + .05)}{.20 - .05} = \$700$$

Once again, we can restate the above DCF formula in terms of a multiple of firm cash flows, i.e.,

$$\text{Value} = \text{Cash Flow}_0 \times \left(\frac{[1 + \text{Growth Rate } (g)]}{\text{Discount Rate } (k) - \text{Growth Rate } (g)} \right) \quad (3)$$

where the term in parentheses is the valuation multiple that reflects a growing stream of cash flows. Again, substituting into Equation 3, we get the following:

$$\text{Value} = \$100 \times \left(\frac{1 + .05}{.20 - .05} \right) = \$100 \times 7 = \$700$$

The valuation multiple in this example is equal to the ratio of one plus the growth rate in future cash flows, divided by the difference between the discount rate and the anticipated rate of growth in future cash flows. In the example of the growing perpetuity calculated above, the valuation multiple is 7. Because the multiple is equal to 7, the cap rate is 1/7 or 14.29%, which is less than the 20% discount rate.

Using the cap rate, we value the growing cash flow stream as follows:

$$\begin{aligned} \text{Value} &= \text{Cash Flow}_0 \times \left(\frac{[1 + \text{Growth Rate } (g)]}{\text{Discount Rate } (k) - \text{Growth Rate } (g)} \right) \\ &= \text{Cash Flow}_0 \left(\frac{1}{\text{Cap Rate}} \right) = \$100 \times \left(\frac{1}{.1429} \right) = \$700 \end{aligned}$$

The difference between the discount rate and the capitalization rate increases with the growth rate anticipated in future cash flows. Table 1 illustrates this situation: Panel a includes the valuation multiples corresponding to discount rates ranging from 5% to 22.5% and growth rates ranging from 0% to 5%.³ Note that only when there is zero anticipated growth in future cash flows (as seen in the first column) are the discount rate and the capitalization rate equal to one another. Moreover, as the rate of growth

³ We include only one case where the discount rate of 5% is equal to the growth rate of 5%. For this case, and whenever the growth rate exceeds the discount rate, the Gordon model “blows up” and indicates an infinite value for the investment opportunity. The Gordon model cannot be applied, because the growth rate in cash flows would dominate the effects of discounting. Clearly, this makes little economic sense. In practice, while growth rates may exceed discount rates for temporary periods of very high growth, they will never be greater than the discount rate on a permanent basis, as such a growth rate is not sustainable. The investment would attract new capital, and competition among new entrants would drive down profit margins and consequently growth rates in cash flows.

Table 1 Capitalization Rates, Discount Rates, Growth Rates, and Valuation Multiples

Panel a. Valuation Multiple = $\left(\frac{1+g}{k-g}\right)$						
Growth Rates (g)						
Discount Rates (k)	0%	1%	2%	3%	4%	5%
5.0%	20.00	25.25	34.00	51.50	104.00	infinite
7.5%	13.33	15.54	18.55	22.89	29.71	42.00
10.0%	10.00	11.22	12.75	14.71	17.33	21.00
12.5%	8.00	8.78	9.71	10.84	12.24	14.00
15.0%	6.67	7.21	7.85	8.58	9.45	10.50
17.5%	5.71	6.12	6.58	7.10	7.70	8.40
20.0%	5.00	5.32	5.67	6.06	6.50	7.00
22.5%	4.44	4.70	4.98	5.28	5.62	6.00
Panel b. Implied Capitalization Rate = $1 / \left(\frac{1+g}{k-g}\right) = \left(\frac{k-g}{1+g}\right)$						
Growth Rates (g)						
Discount Rates (k)	0%	1%	2%	3%	4%	5%
5.0%	5.00%	3.96%	2.94%	1.94%	0.96%	NM
7.5%	7.50%	6.44%	5.39%	4.37%	3.37%	2.38%
10.0%	10.00%	8.91%	7.84%	6.80%	5.77%	4.76%
12.5%	12.50%	11.39%	10.29%	9.22%	8.17%	7.14%
15.0%	15.00%	13.86%	12.75%	11.65%	10.58%	9.52%
17.5%	17.50%	16.34%	15.20%	14.08%	12.98%	11.90%
20.0%	20.00%	18.81%	17.65%	16.50%	15.38%	14.29%
22.5%	22.50%	21.29%	20.10%	18.93%	17.79%	16.67%
NM = not meaningful						

increases, the divergence between the cap rate and discount rate widens. In fact, the cap rate is approximately equal to the difference between the discount rate (k) and the growth rate (g), but not quite, because the growth rate also appears in the numerator of the valuation multiple, i.e., $(1+g)/(k-g)$.⁴

⁴ It is also common to define the Gordon growth model using the projected cash flow for the end of the first period [i.e., $\text{Cash Flow}_1 = \text{Cash Flow}_0(1+g)$], in which case the valuation multiple becomes $\left(\frac{1}{k-g}\right)$ and the capitalization or cap rate equals $(k-g)$.

Office-Building Example

To demonstrate the valuation of an investment with market multiples, let's consider an office building that is currently generating cash flows of \$3 million per year. To value the building using the DCF approach, you need estimates of both the expected growth rates of the cash flows (g) as well as the appropriate discount rate (k), neither of which is readily observable. In contrast, the cash flow multiples approach requires only one input, which can be inferred from previous transactions. For example, if a similar office building that generated \$4 million per year recently sold for \$40 million, one could say that the market multiple is 10 and the cap rate is 10%. If the office building in question is truly comparable, you might then say that its value is \$3 million times 10, or \$30 million. This, of course, is a trivial exercise if the buildings really are directly comparable. The challenge lies in determining the extent to which the buildings are really comparable and in then assessing how their differences should influence the appropriate multiples.

Valuation When the Buildings Are Not Identical In practice, no two investments are identical, so you must assess the extent to which the differences are likely to have a material effect on the valuation multiples. To see how small differences between the comps and the investment being valued can be very important, consider the valuation of the two office buildings described below:

Valuation Analysis				
	Per Square Foot		Total Building	
	Bldg. A	Bldg. B	Bldg. A	Bldg. B
Building size	—	—	50,000 sq. ft.	50,000 sq. ft.
Rent	\$ 30/sq. ft.	\$ 21/sq. ft.	\$ 1,500,000	\$ 1,050,000
Maintenance	(10.00)/sq. ft.	(10.00)/sq. ft.	(500,000)	(500,000)
Net operating income	\$ 20.00/sq. ft.	\$ 11.00/sq. ft.	\$ 1,000,000	\$ 550,000
Selling price information				
Sales multiple for NOI	10/sq. ft.	?/sq. ft.	10	?
Cap rate = 1/Sales multiple	10%/sq. ft.	?/sq. ft.	10%	?
Estimated property value	\$ 200/sq. ft.	?/sq. ft.	\$ 10,000,000	?

Building A sold for \$200 per square foot, or \$10,000,000, which reflects a sales multiple of 10 times the building's \$1,000,000 annual net operating income (NOI) and a capitalization rate of 10%. The two buildings are identical in size (50,000 square feet) and maintenance costs (\$10/sq. ft., or \$500,000 per year), but there is one very important difference. The rental rates (\$30 versus \$21 per square foot) are very different, reflecting their different locations. Building A is in the heart of the downtown business district; Building B is not. However, the local economic factors that drive rental rates affect the two buildings in a similar fashion. Thus, the rental income of each building is similar with respect to *risk and expected growth rates*. This is an important assumption, which we use later.

The two buildings and their operating costs are virtually identical. Because the tenants pay their own utilities and day-to-day operating expenses, the maintenance costs of

each building's owner do not vary over time or with swings in occupancy and rental rates. It is tempting to simply apply Building A's NOI multiple (of 10) to Building B, which would produce a valuation of \$110 per square foot, or \$5,500,000 for the building. But is this valuation correct? Should Building B sell for the same multiple and consequently have the same cap rate as Building A?

At first glance, it does appear that the two buildings are very similar; they are the same size, their rental incomes are determined by the same local economic fluctuations, and both have identical maintenance costs. But, as we now discuss, the different locations of the buildings and the attendant impact of these locations on rental rates have a significant effect on the operating risk characteristics of the two buildings. Specifically, because the rents (revenues) are lower for Building B, it has more **operating leverage**, which means that its cash flows are more sensitive to changes in revenue. As a result, the risks and the growth rates of the cash flows of the two buildings are not the same, which in turn implies that the buildings should sell for different multiples of their cash flows.

Operating Leverage and Investment Risk To understand the differences in operating leverage between the two buildings, note that Building A rents for \$30 per square foot and has maintenance costs of \$10 per square foot, while Building B is much closer to breakeven and rents for only \$21 per square foot and has the same \$10-per-square-foot maintenance costs as Building A. What this means is that Building B has higher fixed costs relative to revenues, meaning that it has substantially higher operating leverage than Building A. To illustrate how operating leverage affects risk, look at what happens to the NOI for Building A when its rental revenues increase or decrease by 20%:

Building A			
% Change in revenues	-20%	0%	20%
Revenues	\$1,200,000	\$1,500,000	\$1,800,000
Maintenance (fixed cost)	(500,000)	(500,000)	(500,000)
Net operating income	\$ 700,000	\$1,000,000	\$1,300,000
% Change in revenues	-20%	0%	20%
% Change in NOI	-30%	0%	30%

A decrease in revenues of 20% results in a 30% drop in Building A's NOI. Similarly, when revenues rise by 20%, Building A's NOI rises by 30%. This dynamic reflects the fact that for Building A the ratio of the % Change in NOI divided by the % Change in revenues is 1.5. Note, however, that this particular relationship between the percent change in operating income and revenues is not constant for a particular investment but varies with the level of firm revenues and NOI. In fact, the closer the revenue level is to breakeven (i.e., NOI equal to zero), the larger will be this multiple. The fact that the percent changes in NOI are greater than the corresponding percent changes in revenues is due to the fact that building maintenance costs do not rise proportionately with revenues. Indeed, in this particular case, the maintenance costs are constant.

Now let's look at the relationship between revenues and NOI for Building B: For Building B, a 20% decrease in revenues leads to a 38.18% decrease in its NOI; when revenues rise by 20%, the NOI increases by 38.18%. Hence, Building B's operating income is more sensitive to changes in rental revenues than is Building A's. In short, Building B

Relative Valuation Using Market Comparables

	Building B		
% Change in revenues	−20%	0%	20%
Revenues	\$840,000	\$1,050,000	\$1,260,000
Maintenance (fixed cost)	(500,000)	(500,000)	(500,000)
Net operating income	\$340,000	\$ 550,000	\$ 760,000
% Change in revenues	−20.00%	0.00%	20.00%
% Change in NOI	−30.18%	0.00%	38.18%

has higher operating leverage than Building A, and, as a result, we expect its operating income to be more volatile in response to changes in rental revenues.

Investigating the Determinants of Cash Flows Even though we have established that the two buildings are *not* perfectly comparable, it is still possible to use an adjusted comparable analysis, based on the sale price of Building A, to value Building B. To do this, we need to first get a better feel for what determines the values of the two buildings by digging a bit more deeply into the determinants of their cash flows. Specifically, we can decompose each building's NOI into rental revenues and maintenance costs. In addition, we can think of distinct *revenue multipliers* and *maintenance-cost multipliers* instead of a composite NOI multiplier. This decomposition helps us analyze how each component of NOI influences the values of the two buildings. Equation 4a defines building value based on NOI as the difference between the value of the building's rental revenues and the value of its maintenance costs:

$$\begin{aligned} \text{Building Value} &= \left(\text{NOI} \times \frac{\text{NOI}}{\text{Multiple}} \right) = \left(\frac{\text{Rental Revenues}}{\text{Revenue Multiple}} \times \text{Revenue} \right) \\ &\quad - \left(\frac{\text{Maintenance Costs}}{\text{Maintenance Multiple}} \times \text{Maintenance} \right) \end{aligned} \quad (4a)$$

or, using capitalization rates:

$$\begin{aligned} \frac{\text{Building Value}}{\text{Value}} &= \left(\frac{\text{NOI}}{\text{NOI Cap Rate}} \right) = \left(\frac{\text{Rental Revenues}}{\text{Revenue Cap Rate}} \right) \\ &\quad - \left(\frac{\text{Maintenance Costs}}{\text{Maintenance Cap Rate}} \right) \end{aligned} \quad (4b)$$

We use the cap-rate formulation in our analysis and assume that the rates applicable to Building A's maintenance and revenues are also appropriate for Building B. Because both buildings cost the same to maintain, it seems reasonable to use identical cap rates for the maintenance costs. Moreover, because we have assumed that rent revenues for each building are similar in terms of both growth and risk, a single revenue cap rate is reasonable to apply to the rents from both buildings.

If the ratio of rents to expenses were the same for both buildings (resulting in identical operating leverage), then these assumptions would imply that cap rates for each building would also be identical. However, as we will show, when two buildings differ in their operating leverage, they will generally differ in their overall cap rates, even when

they have the same revenue and maintenance cap rates. The decomposition method we illustrate here allows the analyst to compare buildings that differ only in their operating leverage. If buildings A and B differ on more than this dimension, additional analysis is required.

Because we assume that the maintenance costs are fixed, they are somewhat easier to value and are a logical starting point of our analysis. Recall that the cap rate and the discount rate are the same for the case of a constant perpetuity. We assume that the discount rate (and cap rate) for maintenance, which is assumed to be certain, is the borrowing rate associated with Building A, which is 8%. Thus, the present value of Building A's maintenance costs using Equation 1 is as follows:

$$\begin{array}{l} \text{Value of} \\ \text{Building A's} \\ \text{Maintenance Costs} \end{array} = \frac{\$500,000}{.08} = \$6,250,000$$

Given the \$10,000,000 value of Building A and the above present value of its maintenance costs, the value of the building's revenues must equal \$16,250,000. This amount is the value of the building plus the value of the maintenance costs. This implies a capitalization rate for Building A's annual rental revenues of 9.23% after rounding (because $\$1,500,000/.0923 = \$16,250,000$).

We now use the capitalization rates from Building A to value Building B. Specifically, we will use the 9.23% cap rate inferred from the valuation of Building A's revenues to value the revenues of Building B as follows:

$$\frac{\$1,050,000}{.0923} = \$11,375,000$$

Moreover, we use the 8% cap rate on Building A's maintenance costs to value the costs for Building B as follows:

$$\frac{\$500,000}{.08} = \$6,250,000$$

The resulting valuation of Building B is \$5,125,000 (which we calculate as \$11,375,000 – \$6,250,000). The details of our analysis of Building B are summarized in the table below:

	Per Square Foot		Total	
	Bldg. A	Bldg. B	Bldg. A	Bldg. B
Borrowing rate	8%	8%	8%	8%
Value of maintenance costs	\$ (125.00)	\$ (125.00)	\$ (6,250,000)	\$ (6,250,000)
Implied revenue value	\$ 325.00	\$ 227.50	\$ 16,250,000	\$ 11,375,000
Implied revenue multiple	10.833	10.833	10.833	10.833
Implied revenue cap rate	9.23%	9.23%	9.23%	9.23%
Property value	\$ 200.00	\$ 102.50	\$ 10,000,000	\$ 5,125,000
Implied multiple of NOI	10.000	9.318	10.000	9.318
Implied cap rate of NOI	10.0%	10.732%	10.0%	10.732%

Based on this analysis, we estimate that Building B is actually worth only \$5,125,000 rather than the earlier estimate of \$5,500,000, which was based on a more naïve application of multiples. The lower value reflects the fact that Building B has more operating leverage and is thus riskier, which in turn implies that it must have a higher cap rate.

Key learning points We can draw two important learning points from the commercial real estate valuation exercise that generalize to almost any investment. The first point is that investments that look very similar on the surface can generate cash flows with very different risks and growth rates, and should thus sell for different multiples. As this example illustrates, operating leverage is an important determinant of value and must be accounted for in choosing comps, and when operating leverage is different, it is important to adjust comps for these differences. Finally, the example illustrates the importance of exercising great care and using creative ways to properly apply the method of market comparables. It is very tempting to assume that because the use of market comparables as a valuation tool appears very simple, the analysis *is* simple. The fact is that valuation using market comparables requires the same diligence and care as discounted cash flow analysis.

3 ENTERPRISE VALUATION USING EBITDA MULTIPLES

The most popular approach used by business professionals to estimate a firm's enterprise value involves the use of a multiple of an accounting earnings number commonly called **EBITDA**, which refers to earnings before interest, taxes, depreciation, and amortization. Analysts generally view EBITDA as a crude measure of a firm's cash flow and thus view EBITDA multiples as roughly analogous to the cash flow multiples used in real estate. As our discussion in this section indicates, sometimes the analogy is good and sometimes it is not.

Enterprise Value Versus Firm Value

The enterprise value of a firm is defined as the sum of the values of the firm's interest-bearing debt and its equity minus the firm's cash balance on the date of the valuation. For example, in August 2005, Airgas Inc. (ARG)⁵ had an enterprise value of \$2.96 billion, consisting of the following components:

Elements of the Firm's Balance Sheet	Values
Non-interest-bearing debt	\$ 669,056,000
Interest-bearing debt (short- and long-term)	808,635,000
Common equity (market value = price per share × shares outstanding)	2,180,000,000
Firm value	<u>\$ 3,657,691,000</u>
Less: Non-interest-bearing debt	(669,056,000)
Less: Cash and equivalents	(32,640,000)
Equals: Enterprise value	<u><u>\$ 2,955,995,000</u></u>

⁵ Airgas Inc. and its subsidiaries distribute industrial, medical, and specialty gases and welding, safety, and related products in the United States.

We can calculate the equity value of Airgas Inc. from enterprise value as follows:

$$\text{Owner's Equity} = \text{Enterprise Value} - (\text{Interest-Bearing Debt} - \text{Cash}) \quad (5a)$$

$$= \text{Enterprise Value} - \text{Net Debt}$$

$$\$2.18 \text{ billion} = \$2.96 \text{ billion} - 0.78 \text{ billion} \quad (5b)$$

where the term *net debt* refers to the firm's interest-bearing liabilities less cash.

The Airgas EBITDA Multiple

To calculate the Airgas EBITDA multiple, we solve the following equation:

$$\frac{\text{Enterprise Value}_{\text{Airgas 2005}}}{\text{EBITDA}_{\text{Airgas 2005}}} = \text{EBITDA Multiple} \quad (6)$$

For example, on August 1, 2005, Airgas's EBITDA was \$340 million and its enterprise value was \$2,955,995,000; this results in an EBITDA multiple for Airgas of 8.69. In the next section, we look at how the Airgas EBITDA multiple, along with multiples of similar firms, might be used to value a privately held company.

EBITDA	\$ 10,000,000
Enterprise value = 10.51 × \$10,000,000	105,100,000
Plus: Cash	2,400,000
Less: Interest-bearing debt	(21,000,000)
Equity value	\$ 86,500,000

Example—Valuing a Privately Held Firm

To explore the use of an EBITDA multiple, we consider the valuation of Helix Corporation, a privately owned company operating out of Phoenix, Arizona, that is a potential acquisition candidate. Because it is privately owned, a potential acquirer cannot observe its market value. However, the acquirer can use similar firms that *are* publicly traded to infer a value for Helix by using the appropriate EBITDA valuation ratio.

We first identify comparable firms from which to calculate an appropriate EBITDA multiple and then apply this multiple to Helix's EBITDA to arrive at an initial estimate of the enterprise value of the firm.⁶ The second step in the process involves refining our estimates of both the EBITDA multiple and the firm's EBITDA. This step constitutes a careful assessment of the basic elements of the valuation (EBITDA and the EBITDA multiple or valuation ratio) so as to tailor the analysis to the specific firm that is being valued. We consider each step in the analysis over the next few pages.

Helix is a regional supplier of specialty gases including nitrogen, oxygen, argon, helium, acetylene, carbon dioxide, nitrous oxide, hydrogen, welding gases, purity grades, and application blends. We identify four national firms that constitute the primary competitors to Helix. These include Air Products (APD), Airgas (ARG), Praxair (PX), and

⁶ We are assuming that Helix's EBITDA for the year just ended is known. This would be the case if Helix and Airgas had entered into negotiations.

Table 2 Comparable-Firm EBITDA Multiples for Helix Corporation

(\$ millions)	Air Products	Praxair Inc.	Applied Industrial Technologies Inc.	Airgas	Average
Enterprise value	\$ 16,100	\$ 19,450	\$ 1,080	\$ 2,996	
EBITDA	1,390	1,810	99	340	
EBITDA multiple	11.58	10.75	10.91	8.81	10.51

Applied Industrial Technologies (AIT). Consequently, we begin by identifying the most recent EBITDA multiples for these firms.

Table 2, which describes financial information about the four public firms that are engaged in a business similar to Helix, reveals that the average EBITDA multiple for comparable firms is 10.51. If Helix anticipates earning \$10 million in EBITDA this year, then our initial estimate of the firm's enterprise value is \$10 million $\times$ 10.51 = \$105.1 million. Helix has a cash balance of \$2.4 million and owes interest-bearing debt totaling \$21 million. Consequently, we estimate the value of Helix's equity to be \$86.5 million.

Just as we did earlier with our real estate valuation exercises, we must now engage in the second phase of the valuation process, which involves tailoring the analysis to Helix. In other words, we consider the need to make adjustments to both the EBITDA and the EBITDA multiple used in Step 1. To gain some perspective on the issues that arise in the second step of the valuation process, we must first take a detour and review the relationship between EBITDA and free cash flow.

EBITDA and Firm Free Cash Flow

When we opened our discussion of the EBITDA enterprise-valuation approach, we noted that analysts often think of EBITDA as a rough estimate of cash flow. Although the two measures are related, EBITDA is not the same as free cash flow (FCF). Figure 1 illustrates the difference between EBITDA and a firm's free cash flow, which can help us understand when the EBITDA valuation approach works well and when it does not. The figure shows that EBITDA falls short of the firm's free cash flow by the sum at the bottom of the rightmost column of Figure 1. Specifically, EBITDA is a before-tax measure and does not include expenditures for new capital equipment (CAPEX) and does not account for changes in net working capital (NWC).

Summarizing Figure 1, we see that all the items listed here make EBITDA overstate Firm FCF.

$$\text{EBITDA} = \text{FCF} + (\text{T} \times \text{EBIT} + \text{CAPEX} + \text{NWC}) \quad (7a)$$

Similarly, solving for FCF,

$$\text{FCF} = \text{EBITDA} - (\text{T} \times \text{EBIT} + \text{CAPEX} + \text{NWC}) \quad (7b)$$

What this last expression illustrates is that FCF is often more volatile than EBITDA. The reason is that FCF accounts for new investments in capital equipment (CAPEX)

Figure 1 EBITDA and Cash Flow

Comparing the Calculation of FCF and EBITDA		Difference
Firm Free Cash Flow (FCF)	EBITDA	FCF – EBITDA
EBIT (Earnings before interest and taxes)	EBIT	0
Less: Taxes = $T \times \text{EBIT}$	NA	$-T \times \text{EBIT}$
Plus: Depreciation and amortization	Plus: Depreciation and amortization	0
Less: Capital expenditures (CAPEX)	NA	$-\text{CAPEX}$
Less: Change in net working capital (NWC)	NA	$-\text{NWC}$
Sum: FFCF	EBITDA	$-T \times \text{EBIT} - \text{CAPEX} - \text{NWC}$

and net working capital (NWC), which are discretionary to varying degrees⁷ and vary over the business cycle, rising in good times and contracting in bad times. Thus, in those years when large capital investments are being made, EBITDA significantly exceeds the firm's free cash flow and vice versa.

Why Use EBITDA Multiples Rather than Cash Flow Multiples?

Although EBITDA provides a crude estimate of a firm's cash flows, it does provide a relatively good measure of the before-tax cash flows that are generated by the firm's existing assets. To see this, reexamine Equation 6 and assume that the firm will not be paying taxes and will not be investing and growing. Because the firm will not be growing, it will not experience any changes in working capital. Under these assumptions, FCF will be equal to EBITDA, as can be seen from Equation 7b. We can value these cash flows using Gordon's growth model (Equation 2), but we must recognize that without new investment, the cash flows from the firm's existing business are likely to diminish over time, as competitors enter the firm's market and as its plant and equipment become obsolete.

Remember, however, that because EBITDA measures only the earnings of the firm's assets already in place, it ignores the value of the firm's new investments. This is clearly a disadvantage of this valuation tool. One might think that it would be better to value the earnings potential of the entire firm using an FCF multiple than to value just its assets in place using an EBITDA multiple. However, as we noted earlier, FCF for most firms is very volatile, because it reflects discretionary expenditures for capital investments and working capital that can change dramatically from year to year. In fact, FCF is frequently negative, because capital expenditures often exceed internally generated capital. As a result, FCF multiples are unlikely to be as reliable as multiples based on EBITDA, and for this reason they are not used as often in practice.

One may ask, however, whether it is reasonable to simply ignore the firm's capital expenditures, which is implicitly being done when using EBITDA multiples. The answer

⁷ For example, to the extent that capital expenditures can be delayed or sped up and credit terms allow the firm some discretion over when to pay for short-term credit, these expenditures are under the control of management.

is that we can ignore capital expenditures if we believe that the firm's investments have zero net present values on average. However, while this might be a reasonable assumption for many mature firms, one certainly does not want to ignore growth opportunities that have positive NPVs. Given this, in any valuation that utilizes EBITDA multiples, it is important to take into account differences between the value of the growth opportunities of the firm being valued (that is, the extent to which the firm has positive-NPV investments) and the growth opportunities of the comparison firms.

In summary, EBITDA multiples provide a good valuation tool for businesses in which most of the value comes from a firm's existing assets. For this reason, we see EBITDA multiples being used in practice primarily for the valuation of stable, mature businesses. EBITDA multiples are much less useful for evaluating businesses whose values reflect significant future growth opportunities.

The Effects of Risk and Growth Potential on EBITDA Multiples

As just mentioned, EBITDA reflects the cash flows generated by the assets the firm currently owns and operates, and, in general, these cash flows are likely to decline over time. However, the decline is likely to be less for some businesses than others, and in some businesses we expect to see increases rather than decreases in cash flows. Consequently, when using an EBITDA multiple, one must consider the growth potential of the firm being valued and select a set of comparable firms to estimate the EBITDA valuation ratio that share similar prospects. In addition, one must consider the risk differences between the firm being evaluated and the comparison firms and determine how these risk differences should be reflected in discount rates and, hence, in multiples.

It is common practice to select firms within the same industry as market comparables, because these are likely to be the most similar to the firm being evaluated. However, there can be substantial differences in risk characteristics and growth opportunities among firms within an industry, and EBITDA multiples should be adjusted to reflect these differences. To a large extent, differences in firm risk arise because of variations in operating leverage across firms, and as our commercial real estate example illustrated, differences in operating leverage can arise because of differences in profit margins. In addition, differences can arise because of differences between fixed and variable operating costs. Firms that incur higher levels of fixed operating costs but lower variable costs will experience more volatile swings in profits as their sales rise and fall over the business cycle. Differences in expected growth rates tend to be somewhat more subjective; however, analysts can consider past growth rates as indicators of future growth rates.

Although it is generally quite difficult to make objective adjustments to EBITDA multiples to allow for differences in risk and growth opportunities, it is critical that these considerations enter into the selection of comparable firms. Let's return to our earlier analysis of the appropriate EBITDA multiple for the valuation of Helix Corporation to see how we might make such adjustments. We might, for example, want to compare Helix's growth opportunities and risk with each of the firms in Table 2. The relatively low EBITDA multiple for Airgas may reflect the fact that its risk (as reflected in the firm's cost of capital) is high and/or its growth opportunities are low when compared to those of the other firms. This begs the question as to whether Helix is more like Airgas or the remaining comparison firms. Although we cannot answer this question without more

information about Helix and the comparison firms, this analysis is exactly what is required to get the best possible estimate of the EBITDA valuation multiple for valuing Helix.

Normalizing EBITDA

We argued earlier that the appeal of EBITDA over cash flow relates to its relative stability when compared to FCF. However, EBITDA does vary over time, and any given year's EBITDA may be influenced by idiosyncratic effects that need to be accounted for when using the EBITDA valuation model. For example, in the Helix Corporation example, we used the firm's most recent year's EBITDA of \$10 million to value the firm. The presumption we made was that this EBITDA was not influenced by any non-recurring special event during the past year. If the EBITDA was affected by any such event, it would lead us to place a value on Helix that is not reflective of the firm's future earning power. For example, if the 2005 EBITDA for Helix reflected the results of a onetime transaction with a customer that contributed \$500,000 to EBITDA but is not likely to be repeated in future years, we might want to make a downward adjustment to EBITDA to \$9,500,000. Similarly, if the 2005 EBITDA reflects extraordinary write-offs of \$250,000, then we might want to make an upward adjustment of EBITDA to \$10,250,000.

Adjusting the Valuation Ratio for Liquidity Discounts and Control Premiums

It should be noted that the value calculated in the preceding exercise is not necessarily the price a buyer will be willing to pay. The price the buyer will pay can vary depending upon the type of buyer. Specifically, financial buyers tend to determine the price they are willing to pay based on an analysis of the acquisition in isolation. That is, they look at the business as a single entity. They then forecast the cash flows they think the business can earn and place a value on them. Strategic buyers, in contrast, tend to think of the value of the business in the context of other businesses they own or are thinking of acquiring. As a consequence, strategic buyers often pay more for a business than a purely financial buyer does.

The rationale for the possible discount is that Helix is a privately held firm and the market-based EBITDA multiples we used in our first-pass valuation were based on a sample of publicly held firms. Private companies often sell at a discount to their publicly traded counterparts because they cannot be sold as easily. A discount of 20% to 30% is not unusual in these circumstances, and the discount is typically attributed to the fact that the shares of privately held firms are less liquid (i.e., harder to sell) than those of public firms. Hence, if we apply a 30% discount, our estimate of the enterprise value of Helix is reduced to $\$73,570,000 = 10.51 \times \$10,000,000 (1 - .3)$. In this case, the estimated value of Helix's equity equals \$54,760,000, i.e.,

Enterprise value (revised)	\$73,357,000
Plus: Cash	2,400,000
Less: Interest-bearing debt	(21,000,000)
Equity value	<u>\$54,760,000</u>

If Helix is being acquired by Airgas, which is a publicly traded firm, is the liquidity discount likely to be an issue? Because Airgas is itself a public firm and is unlikely to want to sell Helix, the relevance of a liquidity discount is subject to debate. The actual price that will be paid on this transaction depends on the relative bargaining strengths of the buyer and seller (i.e., how motivated Airgas is to make the acquisition and how motivated Helix's owners are to sell). This in turn depends on the synergies and improvements Airgas expects to achieve when it takes control of Helix.

The Airgas acquisition of Helix might be considered a “strategic acquisition,” which is a fancy way of saying that there are benefits (or synergies) from control. In contrast to purely financial acquisitions, which often require liquidity discounts, strategic acquisitions often feature control premiums, which can enhance the value of an acquisition target by 30% or more. Once again, when acquiring a private firm such as Helix, the amount of any control premium paid will depend upon the relative strength of the bargaining positions of the buyer and seller. The point here is that adjustments for liquidity discounts and control premiums will vary from one situation to another.

4 EQUITY VALUATION USING THE PRICE–EARNINGS MULTIPLE

Up to this point, our focus has been on the valuation of either major investment projects or the enterprise value of an entire firm. For these applications, it is common for financial analysts to use EBITDA multiples along with the DCF approach. In this section, we explore how analysts value a firm's equity.

One approach would be to use EBITDA multiples to estimate enterprise value, as described in the last section, and then subtract the firm's interest-bearing debt to determine the value of its equity. While this approach is used in practice, it is not the main approach that is used when valuing a firm's equity. Instead, equity analysts tend to focus their attention on estimating the earnings of the firms they evaluate and then use the **price-to-earnings (P/E) ratio** to evaluate the price of the common stock.

The price–earnings (P/E) multiple valuation approach is defined in Equation 8 below:

$$\left(\frac{\text{Price per Share for a Comparable Firm}}{\text{Earnings per Share for a Comparable Firm}} \right) \times \text{Earnings per Share for the Firm Being Valued} \\ = \text{Estimated Value of the Firm's Equity} \quad (8)$$

As we just mentioned, the P/E valuation approach is used to estimate the value of *the firm's equity*. It is not used to find the enterprise value, as we did using the EBITDA valuation method.

Example—Valuing ExxonMobil's Chemical Division Using the P/E Method

To illustrate the use of the P/E ratio valuation approach, let's suppose that ExxonMobil (XOM) was considering the sale of its chemical division. Although this is a hypothetical case, it provides a realistic setting in which the use of the P/E multiple valuation approach

might prove useful. If the chemical division is sold as an independent company to the public, ExxonMobil could use an initial public offering (IPO) to sell the new company's shares. Alternatively, the division could be sold to another company. In either instance, ExxonMobil needs to place a value on the division.

In 2004, the chemical division of ExxonMobil was the third largest chemical company in the world, ranked behind BASF AG (BF) and DuPont (DD) based on total revenues.⁸ The chemical division earned \$3.428 billion in 2004, compared to \$1.432 billion in 2003 and \$830 million in 2002. If the company were to consider selling the division via an initial public offering, a key question it would want to answer would be, "How much can we expect to receive for the sale of the equity?" We can get a quick glimpse at the magnitude of this equity value by applying an average P/E ratio of similar firms to the chemical division's earnings. Table 3 contains P/E ratios for seven of the largest chemical companies in the world. They have an average P/E multiple of 14.28, which implies that, as a first approximation, ExxonMobil's chemical division is likely to be worth about \$48.94 billion = 14.28 × \$3.428 billion.

To refine our estimate of the value of ExxonMobil's (XOM's) chemical division, we want to more closely scrutinize the market comparables and evaluate the extent to which they are really similar to ExxonMobil's chemical division. For example, we noted above that the revenues of ExxonMobil's chemical division make it the third largest chemical company in the world. If firm size is an important determinant of P/E ratios, then the appropriate comparison group would consist of the very largest firms from the industry. Based on their market capitalizations, which is equal to price per share multiplied by the number of shares of common stock outstanding (found in Table 4), the four largest firms include BASF AG, Bayer AG, Dow Chemical, and E I DuPont. The average P/E multiple for these four firms is 15.935, and if we apply this multiple to the valuation of ExxonMobil's equity in the chemical division, the

Table 3 Chemical Company Price/Earnings Ratios (August 16, 2005)

	Share Price	÷	EPS	=	P/E Ratio
BASF AG (BF)	\$ 70.47		\$ 5.243		13.44
Bayer AG (BAY)	35.64		1.511		23.59
Dow Chemical (DOW)	47.40		4.401		10.77
E I DuPont (DD)	41.00		2.572		15.94
Eastman Chemical (EMN)	51.69		5.75		8.99
FMC (FMC)	59.52		5.729		10.39
Rohm & Hass (ROH)	45.02		2.678		16.81
			Average		14.28

⁸ Not every firm makes its divisional reporting available in its public filings. Where no divisional reporting is available, this type of analysis would be very difficult to carry out by an outside analyst. However, the company could, as we propose here, perform the analysis internally so long as it maintains divisional financial reports.

Table 4 Market Capitalization and P/E Ratios for the Chemical Industry Comps

	P/E Ratio	Market Capitalization (billions)
BASF AG (BF)	13.44	\$38.25
Bayer AG (BAY)	23.59	25.63
Dow Chemical (DOW)	10.77	45.25
E I DuPont (DD)	15.94	40.61
Eastman Chemical (EMN)	8.99	4.10
FMC (FMC)	10.39	2.20
Rohm & Hass (ROH)	16.81	10.01
Average (Big 4)	15.94	\$37.44
Average (Small 3)	12.06	5.44

estimate to \$54.63 billion. This is a number that ExxonMobil's management would find much more to their liking!

However, there is substantial dispersion in the P/E ratios in this industry. Hence, to value the ExxonMobil chemical division, we must ask whether its risk and growth potential are more like those of Bayer AG, which has a P/E ratio of 23.59, or more like Dow, which has a P/E ratio of only 10.77. A complete answer to this question would require more in-depth analysis of differences and similarities between ExxonMobil's chemical division and those of the comparable firms, which would certainly include an analysis of the growth prospects and the operating leverage of the different firms.

The remainder of our P/E discussion delves into the impact of a firm's growth prospects on its P/E ratio, both in the context of a stable firm that grows at a steady rate forever and that of a high-growth-rate firm that enjoys a period of high rates of growth followed by a reduction to a sustainable but lower rate of growth.

P/E Multiples for Stable-Growth Firms

A stable-growth firm is one that is expected to grow indefinitely at a constant rate. The P/E multiple of such a firm is determined by its constant rate of growth and can be calculated by solving the Gordon growth model, applied to the valuation of a firm's equity and found in Equation 9 below:⁹

$$P_0 = \frac{\text{Dividend per Share}_0(1 + g)}{k - g} = \frac{\text{Earnings per Share}_0(1 - b)(1 + g)}{k - g} \quad (9)$$

⁹ Note that this equation is simply Equation 2 applied to the firm's equity rather than to its enterprise value.

TECHNICAL INSIGHT

Current Versus Forward Earnings and the P/E Ratio

The price–earnings (P/E) ratio is a simple concept: current market price of a firm’s common stock divided by the firm’s annual earnings per share. Although the current market price is an unambiguous variable, earnings are not. For example, the earnings variable sometimes represents earnings per share for the most recent year. In that case, the P/E ratio is referred to as the *current P/E ratio* or *trailing P/E ratio*. There is yet another commonly used definition of the P/E ratio that defines earnings per share by using analysts’ forecasts of the next year’s earnings. This is the *forward P/E ratio*.

To illustrate, let’s consider the sample of chemical companies used in the valuation example of the ExxonMobil chemical division. The current (trailing) and forward P/E ratios for these firms are found below:

	Share Price	Current EPS	Current/ Trailing P/E Ratio	Forecast EPS	Forward P/E Ratio
BASF AG (BF)	\$ 70.47	\$ 5.243	13.44	\$ 7.27	9.69
Bayer AG (BAY)	35.64	1.511	23.59	2.69	13.27
Dow Chemical (DOW)	47.40	4.401	10.77	5.71	8.30
E I DuPont (DD)	41.00	2.572	15.94	3.04	13.48
Eastman Chemical (EMN)	51.69	5.75	8.99	5.93	8.71
FMC (FMC)	59.52	5.729	10.39	5.66	10.51
Rohm & Hass (ROH)	45.02	2.678	16.81	3.12	14.44
		Average	14.28		11.20

Forward P/E ratios are lower than the current (trailing) P/Es in the six instances where earnings are expected to grow and are higher in the one instance (FMC) in which earnings per share is expected to decline. Although the differences in P/E are not dramatic for these sample firms, they can be dramatic in turnaround situations or in the case of firms that are facing large changes in their earnings prospects for the future. For these cases, it is important for the analyst to look beyond the current or trailing P/E when trying to compare the market’s current valuation of a share of stock to that of comparable firms.

where b is the retention ratio, or the fraction of firm earnings that the firm retains, implying that $(1 - b)$ is the fraction of firm earnings paid in dividends, g is the growth rate of these dividends, and k is the required rate of return on the firm’s equity. By rearranging the terms in this equation, we can express the P/E ratio as follows:

$$\text{P/E Ratio} = \frac{(1 - b)(1 + g)}{k - g} \quad (10a)$$

To look more deeply into the determination of P/E ratios, we first observe that firms are able to grow their earnings by reinvesting retained earnings in positive-net-present-value projects. We will assume that these positive-NPV investments earn a rate of return, which we will denote as r , that exceeds the firm's required rate of return, k . Well-positioned firms with competitive advantages, intellectual property, patents, and managerial expertise are able to generate both higher rates of return on new investment, as well as opportunities to reinvest more of their earnings. It is the combination of the amount by which r exceeds k and the fraction of firm earnings that can be profitably reinvested each year ($1 - b$) that determines the firm's P/E ratio.

To illustrate, suppose that a firm with \$100 million of invested capital generates returns on invested capital of 20% per year. If the firm pays out 100% of its earnings each year, then the dividend will be \$20 million each year. In this case, the growth rate of the firm's dividends, g , is zero. Now suppose that the firm pays only 40% of its earnings in dividends and reinvests the remaining 60% in its operations. In this case, the first dividend will be $D_1 = \$100M \times (.2) \times (.4) = \8 million, leaving the firm with a capital base of \$112 million. If the firm continues to generate the same 20% return on the larger capital base, and if it continues to pay out 40% of its earnings and retain 60%, its second-year dividend will be $D_2 = \$112M \times (.2) \times (.4) = \8.96 million. Dividends will continue to grow at $(\$8.96 - \$8.00)/\$8.00 = 12\%$ annually as long as the dividend payout policy continues and the firm's return on capital remains equal to 20%.

Under these assumptions, we can express a firm's dividend growth rate as the product of its retention rate, b , and the rate of return it can provide on newly invested capital, r . We can now restate the P/E multiple from Equation 10a as follows:

$$\text{P/E Multiple} = \frac{(1 - b)(1 + g)}{k - g} = \frac{(1 - b)(1 + br)}{k - br} \quad (10b)$$

Using Equation 10b, we see that for a stable-growth firm, the P/E multiple is determined by the firm's dividend payout policy (i.e., 1 minus the retention ratio, or $1 - b$); the required rate of return of the stockholders, k ; and the rate of return the firm expects to earn on reinvested earnings, r . Thus, we can use Equation 10b to explore the relationship between the P/E multiple and each of these key variables.

To illustrate the effect of differences in anticipated rates of growth and retention ratios on the P/E ratio, let's consider the two firms (A and B) described in Table 5. Firm A has the opportunity to create value for shareholders by retaining and reinvesting its earnings. This is reflected in the fact that the firm's return on invested capital, r , is 10%, which is greater than the stockholders' required rate of return, k , of 8%. Firm B, on the other hand, hurts its shareholders if it reinvests, because its return on invested capital (10%) is less than the stockholders' required rate of return (12%).

Note that for Firm A, the P/E multiple rises as the fraction of its earnings that can be retained and profitably invested rises. This reflects the creation of shareholder wealth that occurs when a firm with an 8% required rate of return earns 10% on its new investment. The P/E multiple found in Equation 10b becomes infinite, however, when the retention ratio hits 80%. At an 80% retention ratio, the firm's growth rate

Table 5 The P/E Multiple and Dividend Policy for a Stable-Growth Firm

Firm A: $r = 10\%$, $k = 8\%$			Firm B: $r = 10\%$, $k = 12\%$		
Retention Ratio (b)	Growth Rate (g)	P/E Multiple	Retention Ratio (b)	Growth Rate (g)	P/E Multiple
0%	0%	12.50	0%	0%	8.33
10%	1%	12.99	10%	1%	8.26
20%	2%	13.60	20%	2%	8.16
30%	3%	14.42	30%	3%	8.01
40%	4%	15.60	40%	4%	7.80
50%	5%	17.50	50%	5%	7.50
60%	6%	21.20	60%	6%	7.07
70%	7%	32.10	70%	7%	6.42
80%	8%	Undefined*	80%	8%	5.40
90%	9%	Undefined*	90%	9%	3.63
100%	10%	Undefined*	100%	10%	0.00

Legend
 r = return on reinvested earnings
 k = stockholders' required rate of return
 * These cells have negative P/E multiples. Here the formula for P/E multiple $(1 - b)(1 + g)/(k - g)$ does not apply, because for $k < g$, the series (perpetuity) does not converge. Intuitively, this situation implies that the earnings are growing faster than the cost of capital forever. This situation cannot arise in a competitive economy, where free entry will drive down high growth rates to the feasible range (where $k < g$).

(which equals the product of b and r) equals 8%. A quick glance at Equation 10b tells us why this happens: The growth rate and required rate of return on equity are both 8%; thus, the value of the firm's equity is undefined. The situation with Firm B is the inverse of that of Firm A: The P/E multiple decreases as the firm retains a larger fraction of its earnings (i.e., as b increases). This, of course, reflects the fact that the firm is able to earn only 10% by reinvesting the earnings when its stockholders demand a 12% return.

P/E Multiple for a High-Growth Firm

We can relate the P/E multiple for a high-growth firm to company fundamentals in much the same way as we did for a stable-growth firm—with one important difference. Because we do not expect a firm to be able to achieve high growth forever, describing the firm's growth prospects requires two growth periods. We assume that the firm experiences very high growth lasting for a period of n years, followed by a

period of much lower but stable growth.¹⁰ Equation 11 captures the valuation of the equity of such a firm:

$$P_0 = \frac{\text{EPS}_0(1 + g_1)(1 - b_1)}{k - g_1} \left(1 - \frac{(1 + g_1)^n}{(1 + k)^n} \right) + \frac{\text{EPS}_0(1 + g_1)^n(1 - b_2)(1 + g_2)}{k - g_2} \left(\frac{1}{(1 + k)^n} \right) \quad (11)$$

In this equation we abbreviate earnings per share₀ as EPS₀, and both the retention ratio (*b*) and the growth rate (*g*) are now subscripted to reflect the fact that the firm's dividend policy and its growth prospects can be different for the two growth periods.¹¹

We can interpret Equation 11 as follows: The first term on the right-hand side expresses the present value of a stream of dividends received over the next *n* years. For the high-growth firm, dividends grow at a rapid rate equal to *g*₁ for *n* years, after which time the *n*th-year dividends grow at a constant (and lower) rate *g*₂ forever.

We can now solve for the P/E multiple simply by dividing both sides of Equation 11 by EPS₀. The result is found below:

$$\frac{P_0}{\text{EPS}_0} = \frac{(1 - b_1)(1 + g_1)}{k - g_1} \left(1 - \frac{(1 + g_1)^n}{(1 + k)^n} \right) + \frac{(1 - b_2)(1 + g_1)^n(1 + g_2)}{k - g_2} \left(\frac{1}{(1 + k)^n} \right) \quad (12)$$

Let's analyze the determinants of the P/E multiple for the high-growth firm. Recalling that *g*₁ = *b*₁*r*₁ and *g*₂ = *b*₂*r*₂, we see that the firm's P/E multiple is a function of the firm's dividend policy in the two periods (i.e., *b*₁ and *b*₂), the return on equity that the firm earns on its invested capital (*r*₁ and *r*₂), and the stockholders' required rate of

¹⁰ You will recall that we are describing a two-stage growth model similar to our previous discussion. The difference here is that we define equity value as the present value of expected future dividends (not equity free cash flow).

¹¹ Earnings would grow at *g*₁ for *n* years and at *g*₂ immediately thereafter. In contrast, dividends would grow at *g*₁ for *n* years and at a lower *g*₂ after (*n* + 1) years. But in the intervening period, (*n*, *n* + 1), there would be a discontinuity in dividend growth, as shown below:

Year	Earnings	Imputed Earnings Growth	Dividends	Imputed Rate of Dividend Growth
(<i>n</i> - 1, <i>n</i>)	EPS ₀ (1 + <i>g</i> ₁) ^{<i>n</i>}	<i>g</i> ₁	EPS ₀ (1 + <i>g</i> ₁) ^{<i>n</i>} (1 - <i>b</i> ₁)	<i>g</i> ₁
(<i>n</i> , <i>n</i> + 1)	EPS ₀ (1 + <i>g</i> ₁) ^{<i>n</i>} (1 + <i>g</i> ₂)	<i>g</i> ₂	EPS ₀ (1 + <i>g</i> ₁) ^{<i>n</i>} (1 + <i>g</i> ₂) (1 - <i>b</i> ₂)	[(1 + <i>g</i> ₂) (1 - <i>b</i> ₂)]/ (1 - <i>b</i> ₁) - 1
(<i>n</i> + 1, <i>n</i> + 2)	EPS ₀ (1 + <i>g</i> ₁) ^{<i>n</i>} (1 + <i>g</i> ₂) ²	<i>g</i> ₂	EPS ₀ (1 + <i>g</i> ₁) ^{<i>n</i>} (1 + <i>g</i> ₂) ² (1 - <i>b</i> ₂)	<i>g</i> ₂

return (k). Again, the relationship between r (both r_1 and r_2) and k —the return on reinvested earnings and the stockholders' required rate of return, respectively—is the key determinant of the P/E multiple. If $r > k$, the firm's P/E multiple increases, and the amount of the increase is determined by its retention ratio, b .

To illustrate the use of the P/E ratio for a high-growth firm, consider Google Inc., which was formed in 1998 and went public in 2004 and in April 23, 2010 its share price was \$454, giving the firm a P/E multiple of 24.7 based on 2009 earnings of \$22.07 per share. To evaluate whether this high P/E ratio can be justified, we will use Equation 12, which provides for two growth periods: a finite period of high growth followed by a stable-growth period.

To analyze Google's P/E ratio using Equation 12, we use information found in Panel a of Table 6, along with estimates of the duration of the high-growth period (n in Equation 12), the dividend payout ratio after year n (i.e., $[1 - b_2]$), and the anticipated rate of growth in earnings (g_2) after year n . Recall that the growth rate in earnings after year n can be estimated as the product of the fraction of the firm's earnings that are retained (i.e., that are not paid out in dividends) and the rate of return earned on reinvested earnings, or the return on equity. Thus, the key determinants of Google's P/E are the length of the high-growth period, in combination with the dividend payout ratio and the return on equity during the post-high-growth period. Table 6 contains three scenarios that include sets of these key parameters, each of which produces the observed P/E of 24.7. Of course, these are not the only possible combinations of these parameters that will produce a P/E ratio of 24.7 for Google. However, they do illustrate the feasibility of obtaining a P/E ratio that is consistent with its observed market P/E.

To determine how one might evaluate the plausibility of these scenarios, let's consider Scenario #1 in more depth. Scenario #1 assumes that Google will be able to reinvest all of its earnings for the next 6 years at its current return on equity of 21.1%. Because the firm retains all of its earnings, the rate of growth in earnings is also 21.1% (recall that the rate of growth is equal to the product of the return on equity and the retention ratio). Thus, in six years, Google's estimated earnings per share will be $\$69.60 = \$22.07(1 + .211)^6$. At this future date, the firm's P/E ratio will be 14.93, which is much lower than its current P/E ratio of 24.7. The reason for the decline in the P/E ratio is that Google's growth rate in this later period drops from 21.1% to only 2.68%. By multiplying this estimate of the P/E ratio in 2018 by the estimate of the future earnings per share, \$80.61, we estimate that Google's stock price will appreciate to \$1,203.38 at the end of the 12-year period of high growth.

Is it really likely that Google will be able to maintain a 21.1% return on its reinvested earnings for six years? If Google can indeed accomplish this, it will earn \$22.14 billion in 2016, which would be almost double Microsoft Corporation's 2009 earnings of \$16.91 billion. One might decide that this particular set of assumptions is overly optimistic and conclude that Google's stock is overpriced. Alternatively, one might consider whether Scenarios #2 and #3 are more plausible. These scenarios make less optimistic assumptions about the duration of the high-growth-rate period but justify Google's current P/E multiple by assuming a higher long-term growth forecast. For example, in Scenario #2, Google's earnings after six years (in 2016) are much lower than under Scenario #1¹² and are less than Microsoft's current earnings, but its P/E ratio is 18.75,

¹² In Scenario #2, Google's EPS are assumed to grow at 24.7% for only five years and then at 8.16% thereafter.

Table 6	Analyzing the P/E Ratio of a High-Growth Firm —Google Inc.									
Panel a. Current Stock Price and Earnings Information										
P/E ratio	74.5	Return on equity	24.7%							
Beta	0.81	Dividend payout	0.0%							
Risk-free rate	5.02%	Cost of equity	9.07%							
Market risk premium	5.00%									
Panel b. Scenario Analysis										
	Growth Period, <i>n</i>	Dividend Payout ($1 - b_2$)	Return on Equity	Growth Rate, <i>g</i> ₂	P/E Ratio	Estimates for 2018				
						EPS	P/E	Stock Price	Net Income	
Scenario #1	12	58.94%	12.00%	4.93%	74.484	\$ 80.61	14.93	\$ 1,203.38	\$24,432,218,360	
Scenario #2	5	31.98%	12.00%	8.16%	74.484	\$ 29.77	38.12	\$ 1,135.04	\$ 9,023,395,570	
Scenario #3	5	58.94%	18.05%	7.41%	74.484	\$ 28.35	38.13	\$ 1,080.86	\$ 8,592,575,505	

which is higher than the 12.80 P/E under Scenario #1. This difference in P/E ratios reflects the fact that in Scenario #2, Google's earnings during the second growth period are expected to grow at 6.64%, compared to 2.68% for Scenario #1.

As this Google example illustrates, the use of P/E multiples, like all of the approaches we consider, provides equity valuations that are only as good as the judgment of the equity analyst who performs the analysis. The P/E-multiples approach provides a useful tool for taking a set of assumptions and translating them into an estimate of stock price. When carrying out a valuation analysis, one should consider a variety of plausible scenarios and determine how sensitive the valuation model's output is to the duration of the period of high growth as well as to the reinvestment rate and return on equity that will prevail after the period of high growth.

5 PRICING AN INITIAL PUBLIC OFFERING

When a firm sells its shares in the public market for the first time, the process of offering those shares is commonly referred to as an *initial public offering*, or *IPO*. As we will briefly discuss in this section, the market comparables approach used in this chapter plays an important role in the pricing of IPOs.

As a first step in the IPO process, the lead underwriter (that is, the investment banker who manages the IPO process for the company that is going public) determines an initial estimate of a range of values for the issuer's shares. The estimate is typically the result of a comparables valuation analysis that utilizes several valuation ratios such as the ones discussed in this chapter. For example, the underwriter may estimate enterprise value using an EBITDA multiple and then subtract the firm's net debt (i.e., interest-bearing debt less the firm's cash reserves) to get an estimate of the issuer's equity value. The price per share, then, is simply equity value divided by the number of shares the firm is issuing. Given the variety of comparables and valuation ratios typically used in this analysis, this exercise will result in a range of equity prices (say, \$10 to \$15 per share) for the new issue, not a single offering price.

After setting the initial price range, the underwriters go through what is known as a "book-building" process, where they gauge the level of investor interest. Specifically, over the weeks leading up to the offering date, the lead underwriter and company executives travel around the country meeting with potential investors. These visits are known as the "road show." During these visits, the underwriter's sales force collects information from potential institutional investors related to their interest in purchasing shares at various prices within the initial valuation range. This information forms the basis for the "book," which contains nonbinding expressions of interest in buying shares of the IPO at various prices, generally within the initial price range.

Finally, during the pricing meeting on the eve of the IPO, the investment banker and company executives meet to decide on the initial offering price. Narrowing the range of prices down to a single price is a result of the judgment of company executives and the following considerations:

- An updated valuation analysis based on that day's pricing information for comparable companies and recent IPO transactions in combination with the most up-to-date measures of company performance (i.e., EBITDA).
- An analysis of the level of interest in the new offering that is determined in the book-building process. Obviously, high expressions of interest will encourage the firm to move the offering price toward the upper end of the initial pricing range and vice versa.

- Finally, the underwriters like to price the IPO at a discount, typically 10% to 25%, to the price the shares are likely to trade on the market. Underwriters argue that this helps generate good after-market support for the offering.

6 OTHER PRACTICAL CONSIDERATIONS

In this section, we close our discussion of relative valuation by delving into some practical issues that arise when performing a valuation exercise using multiples. Specifically, the first two steps of the four-step procedure are the most critical to the overall success of the effort. These steps involve choices that the analyst must make regarding the set of comparable firms and the particular valuation metric used to determine relative value. In addition, we open a discussion dealing with the use of multiples valuation and DCF valuation.

Selecting Comparable Firms

We now circle back to Step 1—choosing comparable firms—to explore it in more detail. Although it is the first task facing the analyst, it is perhaps the most difficult. Comparable firms share similar operating and financial characteristics. In particular, comparable firms should share similar growth prospects and operating cost structures as well as similar capital structures.

The typical approach taken when selecting comparable firms entails using firms from the same industry group. This makes sense for a number of reasons. First, firms within a given industry tend to utilize similar accounting conventions, which allows the analyst to directly compare the various accounting ratios. Second, firms in the same industry tend to have similar risks and growth prospects.¹³ However, as we have discussed, different firms in the same industry often have very different management philosophies, which lead to very different risk and growth profiles. In addition, firms often do business in multiple industries, making it difficult to determine to which industry group a firm belongs. For example, in our earlier analysis of ExxonMobil (XOM), we found that it had three operating divisions located in different industries.¹⁴ When this occurs, the analyst may find it necessary to utilize an average of the industry valuation ratios corresponding to each of the firm's operating business units.

Choosing the Valuation Ratio

Although earnings and EBITDA valuation ratios tend to be the most commonly used, some firms have either zero or negative earnings. When this is the case, analysts tend to look to other valuation ratios for guidance. For example, as we discussed earlier, real

¹³ These results are based on a comparison of observed prices to estimated prices (based on the various ways of forming groups of comparable firms and assessing the significance of the absolute deviation, measured as a percent of the observed price). See A. Alford, "The Effect of the Set of Comparable Firms on the Accuracy of the Price-Earnings Valuation Method," *Journal of Accounting Research* (Spring 1992): 94–108.

¹⁴ These divisions included Upstream—oil and gas exploration and development; Downstream—refining oil and gas for energy use; and Chemicals—conversion of crude into plastics and other nonenergy products.

PRACTITIONER INSIGHT

The Pricing of New Shares in an Initial Public Offering (IPO)—An Interview with J. Douglas Ramsey, Ph.D.*

In 2006, we took EXCO Resources (NYSE: XCO) public with an initial public offering. EXCO had only been a privately held firm because 2003, when we were taken private after being listed on Nasdaq with the help of the private equity firm Cerberus Capital Management LP. The motivation for going public at this time was to raise capital needed to fund some major acquisitions of oil and gas properties and reduce the amount of debt on the balance sheet. This followed on the heels of an equity buyout several months earlier, where certain private equity investors from the 2003 going-private transaction were cashed out. The equity buyout was funded with a combination of debt and new private equity.

The pricing of the new shares is done by the lead investment bank (J.P. Morgan in our case), using market price comparables for similar firms. Specifically, our banker looked at six different ratios equal to enterprise value divided by one of six different valuation metrics that are commonly used in our industry.** This analysis, in combination with the values of EXCO's own valuation metrics, gave us an idea of what type of enterprise value we could expect once the IPO was complete. From this valuation and the number of shares we

planned to sell, we could then impute a price for our shares.

The actual price at which the new shares were offered was not determined until the night before the IPO in a pricing meeting. In this meeting, our investment banker brought an updated analysis of market comparables along with “the book.” The latter is comprised of a listing of the indications of interest in acquiring our shares in the initial offering. This book is built over the weeks prior to the IPO, during what is typically referred to as the “road show,” where company management and representatives of the joint book runners—in our case, J.P. Morgan, Bear Stearns, and Goldman Sachs—barnstorm the country, talking to potential institutional investors about the offering.

*Chief financial officer for EXCO Resources Inc., Dallas, Texas.

**The valuation metrics include company-specific attributes that are believed to be important indicators of the firm's enterprise value. In the case of an E&P company, these typically included such things as the current year's production, proved reserves of oil and natural gas, current-year free cash flow, and estimates of earnings before interest, taxes, depreciation, and amortization plus development and maintenance expenses (EBITDAX) for the next two years.

estate valuations generally include an analysis of the price per square foot of comparable properties. For some businesses, analysts use valuation ratios based on firm sales or book value. Table 7 on the next few pages provides a summary list of some of the most popular valuation ratios. Panel a focuses on equity valuation ratios, and Panel b lists some popular enterprise-valuation ratios.

Maintaining Consistency When Selecting a Valuation Ratio

A simple but important guide for using valuation ratios is that the numerator and denominator of the ratio should be consistent. By consistent we mean that if the price or value metric in the numerator is based on equity value (e.g., price per share), then the denominator should reflect an attribute of the firm that is directly related to share price. For example, net income per share or earnings per share can be directly linked to share price, because earnings represent the income that is available for distribution to shareholders or

Table 7 **Alternative Valuation Ratios**

Panel a. Equity Valuation Ratios			
Valuation Ratio	Definition	Measurement Issues	When to Use
Price to earnings (P/E)	$\frac{\text{Price per Share}}{\text{Earnings per Share}}$ <i>Price per share</i> = Market price per share of common stock <i>Earnings per share</i> = Annual net income ÷ Shares outstanding	<i>Price per share</i>—Typically, the most recent share price is used. However, if share price is very volatile, a normalized price (e.g., an average of beginning and ending prices for the month) may be used. <i>Earnings per share</i>—Although current-year or annualized current-quarter (trailing) earnings can be used, it is not uncommon to use analysts' expectations of future (leading) earnings. Moreover, earnings are typically measured before extraordinary items and may include only earnings from the firm's primary operations.	Firms with an established record of positive earnings that do not have significant noncash expenditures.
Price to earnings to growth (PEG)	$\frac{\text{Price per Share}}{\frac{\text{Earnings per Share (EPS)}}{\div \text{Growth rate in EPS}}}$ <i>Growth rate in EPS</i> = Expected rate of growth in EPS over the next year	<i>Growth rate in EPS</i>—Estimates of growth rates can be made from historical earnings estimates or can be obtained from analysts' estimates.	Firms that face stable prospects for future growth in EPS as well as similar capital structures and similar industry risk attributes.
			Estimated Stock Price = $(\text{EPS})_{\text{Firm}}$ $\times \left(\frac{\text{P/E}}{\text{Ratio}} \right)_{\text{Industry}}$ Estimated Stock Price = $\left(\frac{\text{Growth Rate in EPS}}{\text{EPS}} \right)_{\text{Firm}}$ $\times \left(\frac{\text{PEG}}{\text{Ratio}} \right)_{\text{Industry}}$

Market to book value of equity	Market Value of Equity Book Value of Equity Market value of equity = Price per share $\times$ Shares outstanding Book value of equity = Total assets – Total liabilities	Market value of equity—The same issues that arise in selecting a price per share apply here. Book value of equity—Although the book value of equity is easy to pull from the firm's balance sheet, differences in the ages of firm assets (when acquired and rates of depreciation) as well as other differences in how various assets are accounted for (fair market value—market to market—or cost) and the conservatism employed in accounting practices can lead to variation across firms.	Firms whose balance sheets are reasonable reflections of the market values of their assets. Financial institutions are the classic case.	$\text{Estimated Equity Value} = \left(\frac{\text{Book Value of Equity}}{\text{Market to Book Ratio}} \right)_{\text{Firm}} \times \left(\frac{\text{Market to Book Ratio}}{\text{Book Ratio}} \right)_{\text{Industry}}$
Panel b. Enterprise-Valuation Ratios				
Enterprise value (EV) to EBITDA	Enterprise Value EBITDA Enterprise value = Price per share $\times$ Shares outstanding plus Interest-bearing (short- and long-term) debt less cash EBITDA = Earnings before interest, taxes, depreciation, and amortization	Enterprise value—EV is typically estimated as the market value of the firm's equity (price per share times shares outstanding) plus the book value of the firm's interest-bearing debt. Consequently, the problems that arise in determining which stock price to use (most recent versus normalized) arise here too. EBITDA—This earnings figure is easily accessed from the firm's income statement. However, cyclical variations in earnings and unusual variations in firm revenues may require some normalization to better reflect the firm's future earnings potential.	Firms that have significant noncash expenses (i.e., depreciation and amortization). Examples include industries with large investments in fixed assets, including health care, oil and gas equipment services, and telecommunications.	$\text{Estimated Enterprise Value} = \left(\frac{\text{EBITDA}}{\text{EV to EBITDA Ratio}} \right)_{\text{Firm}} \times \left(\frac{\text{EV to EBITDA Ratio}}{\text{Ratio}} \right)_{\text{Industry}}$

(continued)

Table 7 continued

Valuation Ratio	Definition	Measurement Issues	When to Use	Valuation Model
Enterprise value (EV) to cash flow	$\frac{\text{Enterprise Value}}{\text{Cash Flow per Share}}$ $\text{Enterprise value} = \text{Price per share} \times \text{Shares outstanding plus Interest-bearing debt}$ $\text{Cash flow} = \text{Firm free cash flow (FCF)}$	<i>Cash flow</i> — FCF adjusts EBITDA to consider taxes and additional investment required in capital equipment (CAPEX). Consequently, the issues that arise with FCF include those applicable to EBITDA, plus any normalization of CAPEX required to adjust for extraordinary circumstances in any given year.	Firms with stable growth and thus predictable capital expenditures. Examples include chemicals, paper products, forestry, and industrial metals.	$\text{Estimated Enterprise Value}_{\text{Firm}} = \left(\frac{\text{EV to FCF}}{\text{Ratio}} \right)_{\text{Industry}}$
Enterprise value (EV) to sales	$\frac{\text{Enterprise Value}}{\text{Sales}}$ $\text{Sales} = \text{Firm annual revenues}$	<i>Sales</i> — Firm revenues sit at the top of the firm's income statement. However, they too may not be reflective of the firm's earning potential if they are abnormally large or depressed due to nonrecurring factors.	Young firms and start-ups that do not have an established history of earnings.	$\text{Estimated Enterprise Value}_{\text{Firm}} = \left(\frac{\text{EV to Sales}}{\text{Ratio}} \right)_{\text{Industry}}$

reinvested on their behalf. However, on occasion we have seen analysts report share price relative to EBITDA and share price relative to sales ratios (i.e., price per share divided by sales per share), which are inconsistent ratios, because EBITDA and sales are generated by the entire assets of the firm, not just the equity portion of the firm's balance sheet. Consequently, if one does want to compare firms on the basis of a ratio that involves a variable such as sales per share, one must be careful to select comparable firms that share similar operating leverage and financial leverage with the evaluated firm.

Dealing with Unreliable Financial Information

In any given year, a firm's reported earnings often provide an imperfect measure of the business's ability to generate future cash flows. This is particularly true for young firms that are very far from reaching their potential (i.e., have significant growth potential) or, for that matter, almost any firm going through a transition. Under these circumstances, EBITDA or earnings per share are often negative, making it impossible to rely on ratios based on earnings. In other cases, either the accounting is suspect or the business is able to generate value to its owners, perhaps by paying high salaries, in ways that do not show up in the firm's income statements.

For situations like these, the firm's reported earnings are unreliable indicators of the firm's value. In these circumstances, the analyst can take one of two possible courses of action. The first involves the use of other valuation ratios that may utilize more reliable measures of the firm's ability to generate future cash flows. Most of these alternatives are measures of the size of the business and are thus analogous to valuing real estate on a price-per-square-foot basis. The most popular size measure is firm revenues or sales, followed closely by the book value of the firm's assets. For example, in Panel a of Table 7, the market to book value of equity ratio might be used if the objective is estimating a firm's equity value, or the enterprise value to sales ratio might be used to estimate enterprise value. There are also commonly used measures of size that are unique to particular industries. For example, newspapers, magazines, and cable companies are often valued relative to their number of subscribers, and money management firms and other financial institutions are often valued relative to the amount of assets they manage.

The second course of action open to the analyst who finds that corporate earnings are not reflective of the firm's earning potential involves adjusting or normalizing reported earnings so that they provide a more reasonable reflection of the firm's earning potential. For example, when valuing small family-run businesses, it is not uncommon for the owner to pay himself or herself a noncompetitive salary or to take special perquisites such as a company car, as a way of extracting additional tax-deductible value out of the business. When faced with the prospect of valuing such a business, the analyst must make appropriate adjustments to the firm's earnings before using them to value the enterprise. We will not delve into this process in any detail, because there are many excellent books on financial-statement analysis that detail the circumstances giving rise to a need to normalize earnings and how to do it.

Valuation Ratios Versus DCF Analysis

When valuing major investments, it often makes sense to do both a DCF analysis as well as a valuation that employs a number of different comparables-based multiples. Once the analysis is complete, the analyst is still left with the problem of sorting out the

various value estimates and using his or her professional judgment to arrive at a final valuation. This judgment will depend in part on the quality of information that is available and in part on the purpose of the valuation.

To understand what we mean by the purpose of the valuation, let's consider an investor with an interest in acquiring coal-fired power plants. Suppose that this particular investor is considering the purchase of a specific plant that is quite similar both in age and technology to five other plants that were sold in the last three months, and two others that are currently for sale. In this situation, the quality of the comparables market data is quite good, which means that the acquirer should compare the plants based on valuation ratios, rather than projecting future plant cash flows, estimating an appropriate discount rate, and then performing a DCF analysis. Again, we are assuming that the technologies and the ages of the plants are similar so that their values—relative to their earnings, cash flows, or electricity-generating capacities—should be very similar. In a sense, the question being answered in this valuation is: What should I expect to have to pay for the coal-fired power plant under today's market conditions?

Although the comparables approach provides the appropriate methodology for evaluating whether a particular power plant is favorably priced, it is much less useful for a firm that wants to evaluate whether or not it makes sense to acquire power plants in general. The basic question in this case is: What should a coal-fired power plant be worth today in absolute terms? Note that we are asking a much more basic question here. We no longer just want to know how one coal-fired plant is valued relative to others, but whether coal-fired plants, in general, are good investments. To answer this question, we can still envision comparing EBITDA ratios of power plants and oil refineries as a starting point in the evaluation of which type of investment is likely to be more attractive. However, for this more general question, the analyst should rely much more heavily on DCF analysis, which involves making forecasts of future power output, electricity prices, and costs of generation.¹⁵

7 SUMMARY

In this chapter, we introduced market-based valuation ratios as an alternative to the discounted cash flow (DCF) method for valuing an investment. Although the DCF approach is generally emphasized by academics, practitioners prefer to use market-based multiples based on comparable firms or transactions for valuing businesses. An important advantage of using valuation ratios is that it provides the analyst with a method for estimating the value of an investment without making explicit estimates of either the investment's future cash flows or the discount rate. However, an even more important reason for using valuation comparables is that it makes direct use of observed market pricing information. Intuitively, it does not make sense to use rough estimates of the discount rate and future cash flows to come up with an estimate of the market value of an investment if you can directly observe how the market values such an investment with comparable transactions.

The reality, however, is that using comparable transactions is not as easy as it might at first seem. There are two important steps to the application, and each is critical.

¹⁵ Should the company also mine its coal, the health care liability related to black lung disease can be a very important consideration.

- First, the analyst must identify a set of comparable transactions for which market pricing data is available. The principal weakness of market-based comparables valuation is that we almost never have truly comparable transactions. Investments typically have unique attributes, and in most cases, the investments that are used as comparables are somewhat of a stretch. For this reason, it is critical that we think carefully when determining the appropriate multiples to use. For example, we should use higher multiples when the cash flows of an investment are likely to grow faster than those of the comparable investments, and we should use lower multiples when the risk of the investment is higher. In other words, you should use the insights learned from DCF valuation to augment your implementation of the multiples approach.
- The second critical step in using market comparables entails the selection of an appropriate valuation ratio or metric. By appropriate, we mean that the value of the investment can be thought of simply as a scaled value of the metric where the investment attribute (e.g., earnings, cash flow, square footage) is the scaling variable. For example, when the P/E ratio is used as the valuation metric, we are assuming that the value of the firm's equity is equal to firm earnings scaled by the ratio of share price to earnings. Or, if we're valuing an entire firm, we might use the enterprise value to EBITDA multiple, whereby we scale the firm's EBITDA by the multiple to get an estimate of enterprise value.

PROBLEMS

1 VALUING RESIDENTIAL REAL ESTATE Sarah Fluggel is considering the purchase of a home located at 2121 Tarter Circle in Frisco, Texas. The home has 3,000 square feet of heated and cooled living area, and the current owners are asking a price of \$375,000 for it.

- a. Use the information provided in the following table to determine an initial estimate of the value of the home Sarah is considering:

	Comp #1	Comp #2
Sale price	#240,000	\$265,000
Square footage	2,240	2,145
Selling price/sq. ft.	\$107.14	\$123.54
Time on the market	61 days	32 days

- b. After making her initial estimate of the value of the home, Sarah decided to investigate whether the owner's asking price of \$375,000 might be justified based on unique attributes of the home. What types of details might you recommend Sarah look for in trying to justify the price of the home?
- c. What if the house Sarah is considering had an asking price of \$315,000? What would you recommend Sarah do then?

Relative Valuation Using Market Comparables

2 VALUING COMMERCIAL REAL ESTATE BuildingOne Properties is a limited partnership formed with the express purpose of investing in commercial real estate. The firm is currently considering the acquisition of an office building that we refer to simply as Building B. Building B is very similar to Building A, which recently sold for \$36,960,000.

BuildingOne has gathered general information about the two buildings, including valuation information for Building A:

	Per Square Foot		Total Square Footage	
	A	B	A	B
Building size (sq. ft.)			80,000	90,000
Rent	\$100/sq. ft.	\$120/sq. ft.	\$8,000,000	\$10,800,000
Maintenance (fixed cost)	(23)/sq. ft.	(30)/sq. ft.	(1,840,000)	(2,700,000)
Net operating income	\$ 77/sq. ft.	\$ 90/sq. ft.	\$6,160,000	\$ 8,100,000

Buildings A and B are similar in size (80,000 and 90,000 square feet, respectively). However, the two buildings differ both in maintenance costs (\$23 and \$30 per square foot) and rental rates (\$100 versus \$120 per square foot). At this point, we do not know why these differences exist. Nonetheless, the differences are real and should somehow be “accounted for” in the analysis of the value of Building B using data based on the sale of Building A.

Building A sold for \$462 per square foot, or \$36,960,000. This reflects a sales multiple of six times the building’s net operating income (NOI) of \$6,160,000 per year and a capitalization rate of 16.67%.

- a. Using the multiple of operating income, determine what value BuildingOne should place on Building B.
- b. If the risk-free rate of interest is 5.5% and the building maintenance costs are known with a high degree of certainty, what value should BuildingOne place on Building B’s maintenance costs? How much value should BuildingOne place on Building B’s revenues and, consequently, on the firm?

3 VALUING A PRIVATELY HELD FIRM The auto parts business has three large publicly held firms: O’Reilly Automotive Inc. (Orly), Advance Auto Parts Inc. (AAP), and Auto Zone Inc. (AZO). In addition to these publicly held firms, Carquest is the largest privately held firm in the industry. Assume that in the summer of 2010, your investment banking firm is considering whether to approach the top management of O’Reilly with a proposal that they consider acquiring Carquest, which owns over 4,000 auto parts outlets in the United States. As a preliminary step in the evaluation of the possible acquisition, you have assembled a team of analysts to prepare a preliminary analysis of acquisition price multiples that might be warranted based upon current market conditions.

Relative Valuation Using Market Comparables

The analyst team went to work immediately and compiled the following set of financial information and potential valuation ratios:

Financial Information (millions)	O'Reilly	Advance	Auto Zone
Revenues	\$ 2,120.00	\$ 4,400.00	\$ 5,890.00
EBITDA	321.86	544.38	1,130.00
Net income	171.62	240.16	562.44
Earnings per share	1,507	2.183	7.301
Interest-bearing debt	120.00	560.00	1,720.00
Common equity	1,145.77	939.51	641.16
Total assets	1,713.90	2,615.73	4,401.85
Financial Ratios			
Debt to equity	10.5%	59.6%	269.3%
Gross margins	44.0%	47.3%	49.1%
Operating margins	12.47%	9.42%	16.77%
Expected growth in EPS (5 yrs.)	18.5%	16.0%	13.0%
Market Valuations (millions)			
Market capitalization	\$ 3,240	\$ 3,040	\$ 6,290
Enterprise value	3,360	3,600	8,010
Valuation Ratios			
Enterprise value/EBITDA	10.44	6.61	7.09
P/E ratio (trailing)	19.42	13.30	11.56
P/E ratio (forward)	15.24	11.21	10.21
Beta	1.24	1.79	1.25

- a. How would you use this information to evaluate a potential offer to acquire Carquest's equity?
- b. What do you think is driving the rather dramatic differences in the valuation ratios of the three firms?

4 **NORMALIZING EBITDA** Jason Kidwell is considering whether or not to acquire a local toy manufacturing company, Toys'n Things Inc. The company's annual income statements for the last three years are as follows:

	2010	2009	2008
Revenues	\$ 2,243,155	\$ 2,001,501	\$ 2,115,002
Cost of goods sold	(1,458,051)	(1,300,976)	(1,374,751)
Gross profits	\$ 785,104	\$ 700,525	\$ 740,251
Depreciation and administrative expenses	(574,316)	(550,150)	(561,500)
Net operating income	\$ 210,789	\$ 150,375	\$ 178,751

Relative Valuation Using Market Comparables

- a. Jason has learned that small private companies such as this one typically sell for EBITDA multiples of three to four times. Depreciation expense equals \$50,000 per year. What value would you recommend Jason put on the company?
- b. The current owner of Toys'n Things indicated to Jason that he would not take less than five times 2010 EBITDA to sell out. Jason decides that, based on what he knows about the company, the price could not be justified. However, upon further investigation, Jason learns that the owner's wife is paid \$100,000 a year for administrative services that Jason thinks could be done by a \$50,000-per-year assistant. Moreover, the owner pays himself a salary of \$250,000 per year to run the business, which Jason thinks is at least \$50,000 too high based on the demands of the business. In addition, Jason thinks that by outsourcing raw materials to Asia, he can reduce the firm's cost of goods sold by 10%. After making adjustments for excessive salaries, what value should Jason place on the business? Can Jason justify the value the owner is placing on the business?

5 OPERATING LEVERAGE AND FLUCTUATIONS IN OPERATING EARNINGS Assume that Jason Kidwell (from Problem 4) is able to purchase Toys'n Things Inc., for \$2.2 million. Jason estimates that after initiating his changes in the company's operations (i.e., the salary savings plus outsourcing savings described in Problem 4), the firm's cost of goods sold are 55% of firm revenues and operating expenses are equal to a fixed component of \$250,000 plus a variable cost component equal to 10% of revenues.

- a. Under the above circumstances, estimate the firm's net operating income for revenue levels of \$1 million, \$2 million, and \$4 million. What is the percent change in operating income if revenues go from \$2 million to \$4 million? What is the percent change in operating income if revenues change from \$2 million to \$1 million?
- b. Assume now that Jason is able to modify the firm's cost structure such that the fixed component of operating expenses declines to \$50,000 per year but the variable cost rises to 30% of firm revenues. Answer part a above under this revised cost structure. Which of the two cost structures generates the highest level of operating leverage? What should be the effect of the change in cost structure on the firm's equity beta?

6 VALUING THE EQUITY OF A STABLE-GROWTH FIRM The Emerson Electric Company (EMR) was founded in 1890 and is located in St. Louis, Missouri. The firm provides product technologies and engineering services for industrial, commercial, and consumer markets worldwide. The firm operates in five business segments: process management, industrial automation, network power, climate technologies, and appliance and tools.

The company has a lengthy history of dividend payments and steady growth. In recent years, the firm's dividend payout has averaged 40% of earnings. For 2007, firm earnings were estimated to be \$5.69 a share, and on December 7, 2006, Emerson's shares were trading for \$86.01, which represents a price-earnings ratio of 19.276. Data for the industry, sector, Emerson, and four competitor firms are shown on the next page.

- a. Is Emerson's current stock price reasonable in light of its sector, industry, and comparison firms?
- b. Emerson's beta coefficient is 1.27. Assuming a risk-free rate of 5.02% and a market risk premium of 5%, what is your estimate of the required rate of return for Emerson's stock using the CAPM? What rate of growth in earnings is consistent with Emerson's policy of paying out 40% of earnings in dividends and the firm's historical return on equity? Using your estimated growth rate, what is the value of

Comparable Firms								
Description	Market Cap	P/E	Return on Equity %	Dividend Yield %	Long-Term Debt to Equity	Price to Book Value	Net Profit Margin	Price to Free Cash Flow
Sector: Industrial goods		16.606	14.94%	1.48%	0.87	50.471	5.40%	75.481
Industry: Industrial equipment & components		15.900	18.40%	1.41%	0.649	10.11	7.90%	-134.9
Emerson Electric Co.	\$34.61B	19.276	23.72%	2.40%	0.494	4.257	9.54%	65.156
Parker-Hannifin Corp.	9.81B	14.150	18.16%	1.20%	0.308	2.298	8.25%	34.392
Roper Industries Inc.	4.44B	24.685	14.27%	0.50%	0.603	3.122	11.89%	232.735
Pentair Inc.	3.23B	17.943	11.56%	1.70%	0.485	1.974	4.48%	147.667
Walter Industries Inc.	2.19B	23.537	15.70%	0.30%	4.036	2.731	7.38%	-10.682

Relative Valuation Using Market Comparables

Emerson's shares using the Gordon (single-stage) growth model? Analyze the reasonableness of your estimated value per share using the Gordon model.

- c. Using your analysis in part b above, what growth rate is consistent with Emerson's current share price of \$86.01?

7 HIGH-GROWTH OR STABLE-GROWTH FIRM? Intel Corporation is a leading manufacturer of semiconductor chips. The firm was incorporated in 1968 in Santa Clara, California, and represents one of the greatest success stories of the computer age.

Although Intel continues to grow, the industry in which it operates has matured, so there is some question whether the firm should be evaluated as a high-growth company or stable-growth company from now on. For example, in December of 2007, the firm's shares were trading for \$20.88, which represented a price-earnings ratio of only 17.61. Compared to Google Inc.'s price-earnings ratio of 53.71 on the same date, it would appear that the decision has already been made by the market.

Intel's expected earnings for 2007 were \$1.13 per share, and its payout ratio was 48%. Furthermore, selected financial data for the sector, industry, and seven of the largest firms (including Intel) are found in Exhibit 7.1

- a. Is Intel's current stock price of \$20.88 reasonable in light of its sector, industry, and comparison firms?
- b. Intel has a beta coefficient equal to 1.66. If we assume a risk-free rate of 5.02% and a market risk premium of 5%, what is your estimate of the required rate of return for Intel's stock using the CAPM? What rate of growth in earnings is consistent with Intel's policy of paying out 40% of earnings in dividends and the firm's historical return on equity? Using your estimated growth rate, what is the value of Intel's shares using the Gordon (single-stage) growth model? Analyze the reasonableness of your estimated value per share using the Gordon model.
- c. Using your analysis in part b above, what growth rate is consistent with Intel's current share price of \$20.88?
- d. Analysts expect Intel's earnings to grow at a rate of 12% per year over the next five years. What rate of growth from Year 6 forward (forever) is needed to warrant Intel's current stock price (use your CAPM estimate of the required rate of return on equity)? (*Hint: Use a two-stage growth model where Intel's earnings grow for five years at 12% and from Year 6 forward at a constant rate.*)

8 VALUATION MULTIPLES AND CHANGES IN INTEREST RATES—THOUGHT QUESTION Both EBITDA to enterprise value and P/E ratios can be linked to interest rates through the discount rate used in discounted cash flow valuation. Holding all else equal, when discount rates are higher, valuation ratios are lower. Perhaps because of this, we tend to see stock prices as well as the value of private business transactions decline when interest rates increase.

Macroeconomists like to describe interest rates as consisting of two components: the real interest rate component and an expected inflation component. In some situations, increases in interest rates are the result of an increasing real interest rate; and in other situations, the cause of an interest rate increase is an increase in expected inflation. How might valuation ratios be expected to respond to an interest rate increase generated by an increase in expected inflation versus an interest rate increase that represents an increase in real interest rates?

Exhibit 7.1 Industry Comparables for Intel Corporation

Description	Market Cap	P/E	Return on Equity %	Dividend Yield %	Long-Term Debt to Equity	Price to Book Value	Net Profit Margin	Price to Free Cash Flow
Sector: Technology	5,344.81B	27.716	14.77%	1.90%	0.691	5.588	10.39%	55.435
Industry: Semiconductor—Broad line	252.89B	19.9	16.20%	1.30%	0.096	3.42	15.50%	193.3
Intel Corp.	120.51B	17.622	19.63%	1.90%	0.064	3.437	18.72%	121.039
Texas Instruments Inc.	44.62B	11.08	22.94%	0.50%	0.004	3.71	18.67%	−5,577.55
STMicroelectronics NV	16.35B	24.959	7.81%	0.70%	0.209	1.764	8.24%	−11.219
Advanced Micro Devices Inc.	11.79B	21.152	12.61%	0.00%	0.138	2.088	10.13%	−58.916
Analog Devices Inc.	11.48B	22.667	15.42%	1.90%	NA	3.342	21.48%	311.392
Maxim Integrated Products Inc.	10.28B	23.025	16.93%	1.90%	NA	3.681	21.39%	NA
National Semiconductor Corp.	8.04B	18.49	25.67%	0.60%	0.012	4.481	22.18%	154.483

PROBLEM 9

MINI-CASE PRICING SHARES FOR THE
FRAMCO RESOURCES IPO

Framco Resources is an independent oil and natural gas company that engages in the acquisition, development, and exploitation of onshore North American oil and natural gas properties. The company has followed a strategy of growth through the development of its inventory of drilling locations and exploitation projects, and selectively pursuing acquisitions. The firm's current management team first purchased a significant ownership interest in Framco (which was a public entity) in December 1997 and because then has achieved substantial growth in reserves and production. In 2003, the company was taken private through a buyout financed using debt capital and equity capital provided by a private equity partner. Late in 2005, Framco's board decided that the time was right for the firm to once again become a public entity by doing an initial public offering of its shares.

Framco's board selected an investment banker, who prepared a preliminary analysis of possible offering prices for Framco's shares, found in Exhibits P9.1 and P9.2. The valuation analysis utilizes valuation ratios based on the current enterprise values of five independent oil and gas companies and three key valuation metrics that are commonly used in the industry: estimated reserves; estimated earnings before interest, taxes, depreciation and amortization, and maintenance capital expenditures (EBITDAX); and firm free cash flow.

Exhibit P9.2 contains estimates of Framco's equity and enterprise valuation that would correspond to different IPO share prices. This analysis is based on the assumption that Framco will sell 51.6 million shares of stock for a price of \$20 to \$30 per share. To complete the comparative analysis, Framco's CFO provided the investment banker with the necessary estimates of his firm's proved reserves for 2005, EBITDAX for 2006 and 2007, and free cash flow. These estimates are found below:

Valuation Metric	Framco Estimates
2005E proved reserves	\$700 million
2006E EBITDAX	\$302 million
2007E EBITDAX	\$280 million
Firm free cash flow	\$191 million

- Calculate the valuation ratios found in Exhibit P9.1, using Framco's valuation metrics for each of the alternative IPO prices found in Exhibit P9.2.
- Based on your calculations (and assuming the valuation metrics are used by investors to make value comparisons among independent oil and gas firms), what price do you think Framco's shares will command at the time of the IPO?
- The actual offering price for Framco's shares is not set until the pricing meeting with the investment banker the night before the offering date. At this meeting, the investment banker has not only updated comparables data such as that found in Exhibit P9.1 but also has indications of interest for purchasing the new shares (the book). Additionally, Framco's investment banker reviewed Framco's most recent estimates of the valuation metrics, which were virtually identical to the estimates

Exhibit P9.1
Industry Comparables Used in Framco Resources IPO Pricing Analysis

Company	Firm Characteristics (millions)					Firm Valuation Ratios				
	Enterprise Value	2005E Reserves	2006E EBIT-DAX	2007E EBIT-DAX	Firm Free Cash Flow ¹	2005E Reserves	2006 EBIT-DAX	2007 EBIT-DAX	Firm Free Cash Flow ¹	
Company #1	\$20,547	7,311	\$3,210	\$2,873	\$2,940	\$2.81	6.40	7.15	6.99	
Company #2	21,280	7,220	3,806	3,299	3,502	2.95	5.59	6.45	6.08	
Company #3	4,781	1,411	601	602	450	3.39	7.96	7.94	10.62	
Company #4	2,508	1,222	380	342	252	2.05	6.60	7.33	9.95	
Company #5	2,355	798	460	399	241	2.95	5.12	5.90	9.77	
					Mean	2.83	6.33	6.96	8.68	
					Median	2.95	6.40	7.15	9.77	

¹ Firm free cash flow is calculated in the usual way and includes company estimates of maintenance capital expenditures.

Exhibit P9.2
Enterprise Value Estimates for Framco Based on Alternative IPO Share Prices

	Expected IPO Share Price				
	\$20.00	\$22.00	\$24.00	\$26.00	\$30.00
Diluted shares outstanding (millions)	51.60	51.60	51.60	51.60	51.60
Equity value (millions)	\$1,032	\$1,135	\$1,238	\$1,342	\$1,548
Plus: Net debt (millions) ¹	740	688	637	585	482
Enterprise value (millions)	\$1,772	\$1,824	\$1,875	\$1,927	\$2,030

¹ Net debt equals interest-bearing debt less cash (\$250,000). The interest-bearing debt total declines with increasing IPO share prices, because half the additional proceeds resulting from a higher price are used to retire Framco's debt.

found in Exhibit P9.1. However, the book was quite strong, indicating an oversubscription for the 51.6 million shares at prices at the upper end of the range of prices found in Exhibit P9.2. Should Framco try to raise the offering price outside the original range set forth in Exhibit P9.2? Why or why not?

PROBLEM 10 **MINI-CASE** VALUING GOOGLE IPO SHARE PRICE¹⁶

Google Inc. of Mountain View, California, operates the most popular and powerful search engine on the Web. The company went public using an unconventional Dutch auction method on August 19, 2004. The resulting IPO was the largest Internet IPO ever, raising \$1.67 billion and leaving the firm with 271,219,643 shares of common stock.

While Google commands a wide lead over its competitors in the search engine market, it is witnessing an increased pressure from well-funded rival entities. Yahoo! Inc., with a market cap of approximately \$38.43 billion, is generally regarded as following a business model very similar to Google's.

- a. Using the method of multiples based on both the P/E ratio and the enterprise value to EBITDA ratio, at what price should the stock be offered? Use the data found in Exhibit P10.1 for the following companies as comparables in your analysis: Earthlink, Yahoo, eBay, and Microsoft. Compute the IPO value for Google's shares using each of the comparable firms separately, and then use an average "multiple" of the comparable firms. Use the year-end 2003 balance sheets and income statements of the comparable firms to do the analysis. Assume that Google's forecasted values at the time of the IPO are as follows: Net income is \$400 million, EBITDA is approximately \$800 million, cash and equivalents are \$430 million, and interest-bearing debt (total short-term and long-term) equals only \$10 million.¹⁷

Exhibit P10.1

Financial Information	Earthlink (ELINK)	Yahoo (YHOO)	eBay (EBAY)	Microsoft (MSFT)
2003 Shares outstanding	159,399,000	655,602,000	646,819,000	10,800,000,000
2003 Fiscal close stock price	\$ 10.00	\$ 45.03	\$ 64.61	\$ 25.64
Market capitalization	\$ 1,593,990,000	\$29,521,758,060	\$41,790,975,590	\$276,912,000,000
Short-term debt	\$ 900,000	\$ 0	\$ 2,800,000	\$ 0
Long-term debt	\$ 0	\$ 750,000,000	\$ 124,500,000	\$ 0
Cash & equivalents	\$ 349,740,000	\$ 713,539,000	\$ 1,381,513,000	\$ 6,438,000,000
Short-term investments	\$ 89,088,000	\$ 595,975,000	\$ 340,576,000	\$ 42,610,000,000
EBITDA	\$ 218,100,000	\$ 455,300,000	\$ 818,200,000	\$ 14,656,000,000
Net income	\$ (62,200,000)	\$ 237,900,000	\$ 441,800,000	\$ 9,993,000,000
Calculated EPS	(0.39)	0.36	0.68	0.93

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¹⁷ These estimates for Google are approximate and are based on the prospectus.

- b. Which of the four comparable firms do you think is the best comparison firm for Google? Why?
- c. How has the stock performed because the IPO? Do you believe that Google is currently correctly valued in the stock market? Why or why not?

PROBLEM 11 **MINI-CASE VALUING CONOCOPhillips's ACQUISITION OF BURLINGTON RESOURCES¹⁸**

You have recently been hired as an equity analyst by Wall Street Valuation Consultants and have been assigned the task of valuing the proposed acquisition described in the following press release:

Houston, Texas (December 12, 2005)—ConocoPhillips (NYSE: COP) and Burlington Resources Inc. (NYSE: BR) announced today they have signed a definitive agreement under which ConocoPhillips will acquire Burlington Resources in a transaction valued at \$33.9 billion. The transaction, upon approval by Burlington Resources shareholders, will provide ConocoPhillips with extensive, high quality natural gas exploration and production assets, primarily located in North America. The Burlington Resources portfolio provides a strong complement to ConocoPhillips' global portfolio of integrated exploration, production, refining, and energy transportation operations, thereby positioning the combined company for future growth. (Source: http://www.conocophillips.com/NR/rdonlyres/86E7B7A6-B953-4D0D-9B45-E4F1016DD8FD/0/cop_burlingtonpressrelease.pdf)

In his letter to ConocoPhillips shareholders contained in the company's 2005 annual report, CEO Jim Mulva described the rationale for the proposed Burlington acquisition as follows:

Burlington's near-term production profile is robust and growing, plus Burlington possesses an extensive inventory of prospects and significant land positions in the most promising basins in North America, primarily onshore. With this access to high-quality, long-life reserves, the acquisition enhances our production growth from both conventional and unconventional gas resources.

Specifically, our portfolio will be bolstered by opportunities to enhance production and gain operating synergies in the San Juan Basin of the United States and by an expanded presence and better utilization of our assets in Western Canada. In addition to growth possibilities, these assets also provide significant cash generation potential well into the future.

Beyond adding to production and reserves, Burlington also brings well-recognized technical expertise that, together with ConocoPhillips' existing upstream capabilities, will create a superior organization to capitalize on the expanded asset base. We do not anticipate that the \$33.9 billion acquisition will require asset sales within either ConocoPhillips or Burlington, nor should it change our organic growth plans for the company. We expect to achieve synergies and pretax cost savings of approximately \$375 million annually, after the operations of the two companies are fully integrated.

We anticipate immediate and future cash generation from this transaction that will aid in the rapid reduction of debt incurred for the acquisition and go toward the redeployment of cash into strategic areas of growth. Burlington shareholders will vote on the proposed transaction at a meeting on March 30, 2006. (Source: <http://wh.conocophillips.com/about/reports/ar05/letter.htm>)

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Income Statement and Balance Sheet Values (\$ thousands)					
Ticker	XTO Energy	Chesapeake Energy	Devon Energy	Apache	Burlington Resources
PERIOD ENDING	XTO	CHK	DEVN	APA	BR
	31-Dec-04	31-Dec-04	31-Dec-04	31-Dec-04	31-Dec-04
Income Statement (\$000)					
Total revenue	1,947,601	2,709,268	9,189,000	5,332,577	5,618,000
Cost of revenue	436,998	204,821	1,535,000	946,639	1,040,000
Gross profit	1,510,603	2,504,447	7,654,000	4,385,938	4,578,000
Operating Expenses					
Selling, general, and administrative	165,092	896,290	1,616,000	173,194	215,000
Depreciation, depletion, and amortization	414,341	615,822	2,334,000	1,270,683	1,137,000
Others	11,830	—	—	162,493	640,000
Operating income or loss	919,281	992,335	3,704,000	2,779,568	2,586,000
Income from Continuing Operations					
Total other income/expenses (net)	—	−20,081	64,000	857	—
Earnings before interest and taxes	919,281	972,254	3,768,000	2,780,425	2,586,000
Interest expense	93,661	167,328	475,000	117,342	282,000
Income before tax	825,620	804,926	3,293,000	2,663,083	2,304,000
Income tax expense	317,738	289,771	1,107,000	993,012	777,000
Net income from continuing operations	507,882	515,155	2,186,000	1,670,071	1,527,000
Nonrecurring Events					
Effect of accounting changes	—	—	—	−1,317	—
Net income	507,882	515,155	2,186,000	1,668,757	1,527,000
Preferred stock and other adjustments	—	—	−10,000	−5,680	—
Net income applicable to common shares	\$ 507,882	\$ 515,155	\$ 2,176,000	1,663,074	\$ 1,527,000

Balance Sheet (\$'000)						
Assets						
Current Assets						
Cash and cash equivalents	9,700	6,896	1,152,000	111,093	217,900	
Short-term investments	14,713	51,061	968,000	—	—	
Net receivables	364,836	477,436	1,320,000	1,022,625	994,000	
Inventory	—	32,147	—	157,293	124,000	
Other current assets	47,716	—	143,000	57,771	158,000	
Total current assets	436,965	567,540	3,583,000	1,348,782	3,455,000	
Long-term investments	—	136,912	753,000	—	—	
Property, plant, and equipment	5,624,378	7,444,384	19,346,000	13,860,359	1,103,300	
Goodwill	—	—	5,637,000	189,252	105,400	
Other assets	49,029	95,673	417,000	—	202,000	
Deferred long-term asset charges	—	—	—	104,087	—	
Total assets	6,110,372	8,244,509	29,736,000	15,502,480	15,744,000	
Liabilities						
Current Liabilities						
Accounts payable	425,173	872,539	1,722,000	1,158,181	118,200	
Short/Current long-term debt	75,534	91,414	1,378,000	21,273	2,000	
Other current liabilities	259	—	—	103,487	41,500	
Total current liabilities	500,966	963,953	3,100,000	1,282,891	1,599,000	
Long-term debt	2,053,911	3,076,405	7,796,000	2,619,807	3,887,000	
Other liabilities	199,753	107,395	366,000	1,022,880	851,000	
Deferred long-term liability charges	756,369	933,873	4,800,000	2,372,481	2,396,000	
Total liabilities	3,510,999	5,081,626	16,062,000	7,298,059	8,733,000	

(continued)

Exhibit P11.1
continued

Ticker	XTO Energy	Chesapeake Energy	Devon Energy	Apache	Burlington Resources
	XTO	CHK	DVN	APA	BR
PERIOD ENDING	31-Dec-04	31-Dec-04	31-Dec-04	31-Dec-04	31-Dec-04
Stockholders' Equity					
Preferred stock	—	490,906	1,000	98,387	—
Common stock	3,484	3,169	48,000	209,320	5,000
Retained earnings	1,239,553	262,987	3,693,000	4,017,339	4,163,000
Treasury stock	-24,917	-22,091	—	-97,325	-2,208,000
Capital surplus	1,410,135	2,440,105	9,087,000	4,106,182	3,973,000
Other stockholders' equity	-28,882	-12,193	845,000	-129,482	1,078,000
Total stockholders' equity	2,599,373	3,162,883	13,674,000	8,204,421	7,011,000
Total liabilities and stockholders' equity	6,110,372	8,244,509	29,736,000	15,502,480	15,744,000
Other Financial Data					
Exploration expenses (millions)	\$599.5	\$184.3	\$279.0	\$2,300.0	\$258.0
Shares outstanding (millions)	332.9	253.2	482.0	327.5	392.0
Year-end 2004 closing price	\$35.38	\$16.50	\$38.92	\$50.57	
Market capitalization (millions)	\$11,778.00	\$4,177.80	\$18,759.44	\$16,561.68	
Brief Company Descriptions					
XTO Energy					
XTO Energy Inc., and its subsidiaries, engages in the acquisition, development, exploitation, and exploration of producing oil and gas properties in the United States. The company also produces, processes, markets, and transports oil and natural gas. Its proved reserves are principally located in the Eastern Region, including the East Texas Basin and northwestern Louisiana; Barnett Shale of North Texas; San Juan and Raton Basins of New Mexico and Colorado; Permian and South Texas Region; Mid-Continent and Rocky Mountain Region in Wyoming, Kansas, Oklahoma, and Arkansas; and Middle Ground Shoal Field of Alaska's Cook Inlet. As of December 31, 2005, the company had estimated proved reserves of 6.09 trillion cubic feet of natural gas, 47.4 million barrels of natural gas liquids, and 208.7 million barrels of oil.					
Chesapeake Energy					
Chesapeake Energy Corporation engages in the development, acquisition, production, exploration, and marketing of onshore oil and natural gas properties in the United States. Its properties are located in Oklahoma, Texas, Arkansas, Louisiana, Kansas, Montana, Colorado, North Dakota, New Mexico, West Virginia, Kentucky, Ohio, New York, Maryland, Michigan, Pennsylvania, Tennessee, and Virginia. As of December 31, 2005, the company had proved developed and undeveloped					

reserves of approximately 7,520,690 million cubic feet of gas equivalent, including approximately 103,323 thousand barrels of oil and approximately 6,900,754 million cubic feet of natural gas. It also owned interests in approximately 30,600 producing oil and gas wells, as of the above date.

Devon Energy

Devon Energy Corporation primarily engages in the exploration, development, and production of oil and gas. It owns oil and gas properties located principally in the United States and Canada. The company's U.S. operations are focused in the Permian Basin, the Mid-Continent, the Rocky Mountains, and onshore and offshore Gulf Coast regions; Canadian properties are focused in the western Canadian sedimentary basin in Alberta and British Columbia. Devon Energy also owns properties located in Azerbaijan, China, and Egypt, as well as areas in West Africa, including Equatorial Guinea, Gabon, and Cote d'Ivoire. In addition, the company markets and transports oil, gas, and natural gas liquids; it constructs and operates pipelines, storage and treating facilities, and gas processing plants. As of December 31, 2005, its estimated proved reserves were 2,112 million barrels of oil equivalent. Devon Energy sells its gas production to various customers, including pipelines, utilities, gas marketing firms, industrial users, and local distribution companies.

Apache

Apache Corporation engages in the exploration, development, and production of natural gas, crude oil, and natural gas liquids, primarily in North America. It has exploration and production interests in the Gulf of Mexico, the Gulf Coast, east Texas, the Permian Basin, the spacing of *o* Basin, and the western Sedimentary Basin of Canada. The company also holds exploration and production interests onshore in Egypt, offshore in Western Australia, offshore in the United Kingdom in the North Sea, offshore in the People's Republic of China, and onshore in Argentina. As of December 31, 2005, it had total estimated proved reserves of 976 million barrels of crude oil, condensate, and natural gas liquids, as well as 6.8 trillion cubic feet of natural gas.

However, at an analysts' meeting, CEO Mulva hinted that the price ConocoPhillips paid for Burlington might be viewed as high by some:

In terms of acquisitions, mergers and acquisitions, it really becomes more and more of a seller's market, and terms and conditions are not that attractive to buyers. (Source: <http://news.softpedia.com/news/ConocoPhillips-Plans-To-Acquire-Burlington-14628.shtml>)

Your task is to answer the following basic question: Is Burlington Resources worth the \$35.6 billion offered by ConocoPhillips? Although you are new to the exploration and production (E&P) industry, you have quickly learned that the method of multiples, or market-based comparables, and specifically the ratio of enterprise value (EV) to EBITDAX are typically used as benchmarks to value E&P companies. EBITDAX stands for "earnings before interest, taxes, depreciation and amortization, and exploration expenses." EBITDAX differs from EBITDA in that it adds back exploration expenses in addition to depreciation and amortization—hence the term "EBITDAX."

- a. Using the method of multiples based on enterprise value to EBITDAX, the P/E ratio, and the enterprise value to EBITDA ratio, what should the acquisition price be for Burlington Resources shares? Use the following companies as comparables for your analysis: Chesapeake Energy, XTO Energy, Devon Energy, and Apache. Year-end 2004 balance sheets and income statement summary information as well as market capitalization data are provided in Exhibit P11.1 for Burlington Resources and each of the comparable firms.
- b. Which of the four firms used as comparables do you think is the best comparison firm for Burlington Resources? Why?
- c. Based on your analysis of comparables, did ConocoPhillips pay too much or find a bargain? Explain your answer.
- d. What additional information would help you with this analysis?

PROBLEM 12

MINI-CASE DICK'S SPORTING GOODS IPO¹⁹

Setting: It is early in October 2002 and your investment bank has been hired by the management team at Dick's Sporting Goods (DKS) to determine a valuation/offering price for their stock's initial public offering (IPO).

About Dick's Sporting Goods: DKS, headquartered in Pittsburgh, Pennsylvania, was founded when Dick Stack opened his first store in 1948 with \$300 from his grandmother's cookie jar. This company is growing rapidly and plans to use the IPO proceeds for store openings and acquisitions. DKS operates more than 130 stores in 25 states, primarily in the eastern United States. The store format is a large "big-box" store containing smaller shops featuring sporting goods, apparel, and footwear for activities ranging from football and golf to hunting and camping. Besides brands such as Nike and Adidas, Dick's carries Ativa, Walter Hagen, and others exclusive to the company. At the time of the IPO, the DKS management team included the founder's son, Ed, who holds 37% ownership in the company, and the investment firm Vulcan Ventures (founded by former Microsoft executive Paul Allen), which owns 12% of DKS as a result of a private equity investment.

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Exhibit P12.1

Twelve-Month Indexed Stock Performance of Sporting Goods Retailers

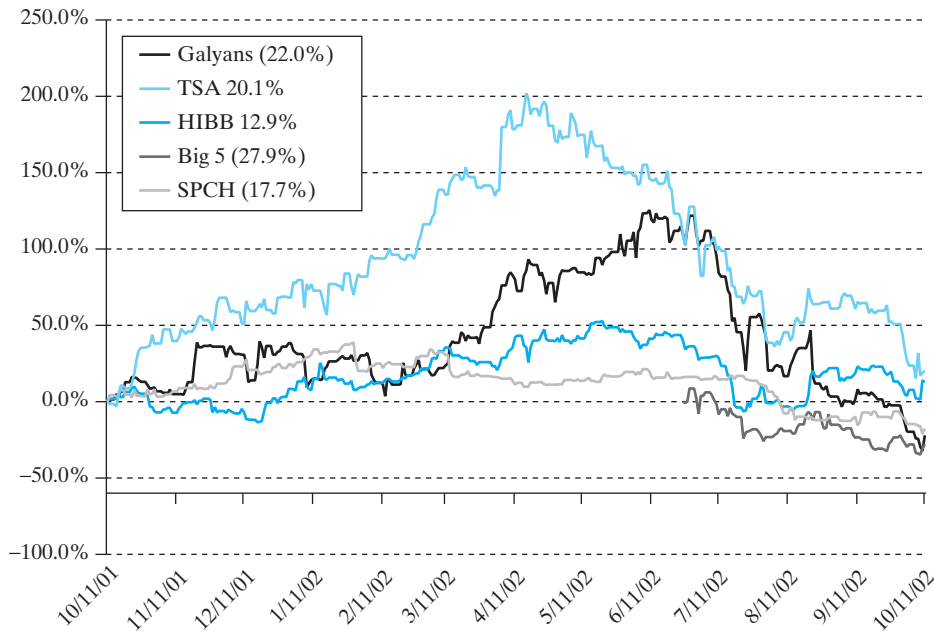


Exhibit P12.2

Twelve-Month Indexed Stock Performance for Sporting Goods Retailer Market Comparable Group Versus the Major Indices

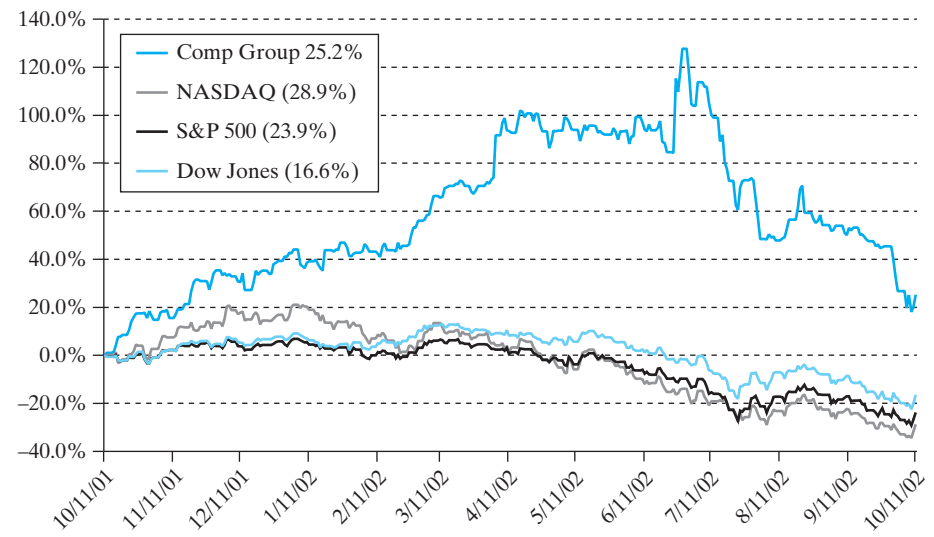


Exhibit P12.3**Dick's Sporting Goods Twelve-Month Performance****Dick's Sporting Goods Financial Data (\$ millions)**

Revenues	\$1,173.794
Gross profit	\$298.453
EBIT	\$55.899
Depreciation and amortization	\$13.499
EBITDA	\$69.398

Balance Sheet Data 8/3/02

Checks drawn	\$33.584
Current portion of long-term debt	\$0.211
Revolving bank line of credit	\$90.299
Long-term debt and capital leases	\$3.466
Total debt	\$127.560
Cash	\$13.874
Stockholders' equity	\$78.984

Industry/Market Overview: With the rise of “big-box” retailers, sporting goods retailers were consolidating. In 1997 and 2001, Gart Sports Company (GRTS) acquired Sport Mart Inc. and OSH Sporting Goods, respectively. Private equity and venture capital firms held equity investments within the industry. DKS was the first retailer to launch an IPO in three months; from the beginning of July 2002 until the time of offering, it had been the lowest combined IPO tally because the second quarter of 1978. In 2002, other retail IPOs had mixed success. Big 5 Sporting Goods' stock price had declined by 23% because its private equity investor, Leonard Green & Partners LP, raised \$105 million in June 2002. Market conditions for equity capital raises were unfavorable, and during the DKS road show, many of the institutional investors expressed concerns that the suggested IPO range of \$15–\$18 per share was too high.

VALUATION ASSIGNMENT

- Based on the DKS financial data (Exhibit P12.3) and sporting goods retailer market comparables valuation (Exhibit P12.4), estimate an implied equity valuation for the firm for its initial public offering. Calculate an implied equity valuation range based on the comparable enterprise-valuation multiples (EV/Revenue, EV/EBIT, EV/EBITDA). Determine an IPO price range for an estimated 9.4 million shares of equity outstanding.
- In your opinion, are the selected publicly traded sporting goods companies shown in Exhibit P12.4 good choices for market comparables for DKS? Why or why not?

Exhibit P12.4

Sporting Goods Retailer Market Comparables Valuation^{20, 21}

	Stock Price	% Change from 12-Month			Calendar EPS ^a		Secular Growth	Calendar P/E Multiples	
		10/9/2002	Low	High	2002E	2003E		2002E	2003E
Sporting Guods Retailers	Ticker								
Galyan's Trading Co.	GLYN	\$ 7.25	0.0%	(68.8%)	\$1.06	\$1.32	23%	6.8×	5.5×
Gart Sports Co.	GRTS	\$14.64	20.8%	(61.2%)	\$1.80	\$2.10	18%	8.1×	7.0×
Hibbett Sporting Goods	HIBB	\$18.89	17.1%	(33.7%)	\$1.35	\$1.62	20%	14.0×	11.7×
Sport Chalet	SPCH	\$ 6.12	0.0%	(43.1%)	\$0.71	\$0.87	14%	8.6×	7.0×
The Sports Authority	TSA	\$ 4.15	1.7%	(71.9%)	\$0.66	\$0.76	13%	6.3×	5.5×
Big 5 Sporting Goods	BGFV	\$ 8.60	0.0%	(39.8%)	\$1.09	\$1.25	15%	7.9×	6.9×
Average			6.6%	(53.1%)			17%	8.6×	7.2×
Median			0.9%	(52.1%)			16%	8.0×	6.9×
Sporting Goods Retailers	LTM Ended	Market Value	Enterprise Value	Enterprise Value/LTM		Debt/Total Cap	LTM Margins		
				Revenue	EBITDA	EBIT	Gross	EBITDA	EBIT
Galyan's Trading Co.	7/02	\$123.50	\$123.50	0.2×	3.0×	4.6×	30%	8%	5%
Gart Sports Co.	7/02	\$178.50	\$313.00	0.3×	6.2×	9.4×	26%	5%	3%
Hibbett Sporting Goods	7/02	\$189.90	\$191.70	0.7×	7.0×	9.2×	31%	10%	8%
Sport Chalet	6/02	\$ 40.40	\$ 45.60	0.2×	4.0×	7.3×	28%	5%	3%
The Sports Authority	7/02	\$136.20	\$331.90	0.2×	4.9×	12.4×	2%	5%	2%
Big 5 Sporting Goods	6/02	\$186.40	\$327.40	0.5×	5.8×	7.0×	35%	9%	7%
Average				0.37×	5.2×	8.3×	25%	7%	5%
Median				0.27×	5.4×	8.3×	29%	7%	4%

^a Annualized 3-year EPS growth rate

Legend: LTM = latest 12 months; Gross (LTM) margin = Gross profit/Revenues; EBITDA (LTM) margin = EBITDA/Revenues; EBIT (LTM) margin = EBIT/Revenues

²⁰ 12-month valuation for the company as of 10/9/02; EPS and growth forecast per First Call analyst consensus.

²¹ Financial data has been normalized for unusual and nonrecurring items.

